I. Cycles through 1 for the $qn + 1$ map: no cycles of length 3, and constraints on longer cycles

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Abstract

For the $qn + 1$ map with q an odd integer at least 3, we prove that no cycle with exactly three distinct odd numbers has starting value 1. This settles the case of length 3 in a conjecture of Santos (2020) that a cycle has starting value 1 only when $q = 5$ or when q is a Mersenne number. We found that for every such q at least 2^{22001}, a lower bound and an upper bound cannot hold at the same time. Bauer and Bennett (2018) give the exponent $\mu \approx 1.61487$, which lies just below the golden ratio, the threshold at which these bounds conflict. Every smaller q is ruled out by a direct computation.

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1 Introduction

This paper studies cycles of the $qn + 1$ map with starting value 1. It is followed by the broader paper [8], which studies cycles of the same map for any starting value.

Fix an odd integer $q \geq 3$. For an odd positive integer n, let v be the largest integer such that 2^v divides $qn + 1$. The map C sends n to

$$C(n) = \frac{qn + 1}{2^v}.$$

The orbit of n is the sequence n, $C(n)$, $C(C(n))$, ... When $q = 3$, C is the odd-to-odd form of the Collatz map, also called the Syracuse function, and the Collatz conjecture, posed by Lothar Collatz in 1937 and still open, asserts that every orbit eventually reaches 1. For $q = 3$, heuristic arguments suggest that almost all orbits eventually reach 1. However, for $q \geq 5$, heuristic arguments suggest that almost all orbits grow without bound and that cycles are rare, although no orbit has been proved to diverge.

We ask for which q a cycle of C has starting value 1. We classify the answers by the length ℓ of the cycle with starting value 1, that is, the number of distinct odd numbers on it.

For $\ell = 1$ the cycle is $1 \to 1$, which happens exactly when q is a Mersenne number (Proposition A.1). For $\ell = 2$, the only q is 5, with $1 \to 3 \to 1$ (Proposition A.2). Besides $1 \to 1$ when q is a Mersenne number, the only known cycle with starting value 1 is $1 \to 3 \to 1$ for $q = 5$.

Santos's conjecture (2020). A cycle of C has starting value 1 only when $q = 5$ or when q is a Mersenne number.

Theorem 1. No odd $q \geq 3$ has a cycle of C of length 3 with starting value 1.

Outline of Theorem 2 (golden-ratio criterion). Fix $\ell \geq 2$, and suppose that, for every integer r with $1 \leq r \leq \ell - 1$, $2^{r/\ell}$ has a restricted irrationality exponent below the golden ratio. Then any q that has a cycle of length ℓ with starting value 1 lies below a bound we can compute. Checking every candidate up to that bound finds every such q, or shows that none exist. The precise statement is in Section 6.

Theorem 3 (finiteness for starting value 1). For each $\ell \geq 2$, only finitely many odd q have a cycle of C of length ℓ with starting value 1.

Theorem 3 is ineffective: it rests on the S-unit form of Ridout's theorem in Section 7, and gives no bound on q. (A finiteness result is called effective if its proof gives an explicit bound on any exception, so that a finite computation can find them all, and ineffective if it gives no such bound.) Theorem 1 is effective and is completed by computation. The proof of Theorem 3, in Section 7, uses only the S-unit form of Ridout's theorem and Theorem 2. Cycles with any starting value are treated in the companion paper [8]. For a cycle of length 2 or 3 with a constant odd starting value, that paper gives a bound on q depending only on the element (Theorem 3), and for Corollaries 12 to 14 the same bound is Corollary 14 of that paper.

2 Definitions and notation

Throughout, $q \geq 3$ is an odd integer and C is the map defined in Section 1.

Mersenne numbers. Let $e \geq 2$ be an integer. A Mersenne number is the integer $2^e - 1$.

Divides. The symbol $|$ means divides, and the symbol $\nmid$ means does not divide. Let a and b be integers, with $a \neq 0$. Write $a \mid b$ when there exists an integer k such that $b = ak$. Write $a \nmid b$ when no such integer k exists.

The 2-adic valuation. For a nonzero integer n, $v_2(n)$ is the largest integer v such that 2^v divides n.

Cycles. A cycle of C of length ℓ consists of ℓ distinct odd positive integers. The starting value is the smallest element x_0. Applying C repeatedly, starting at x_0, produces

$$x_0 \to x_1 \to \cdots \to x_{\ell-1} \to x_0,$$

where the indices are read modulo ℓ.

Starting value 1. A cycle with starting value 1 is a cycle of C of length ℓ with $x_0 = 1$. It is written

$$1 = x_0 \to x_1 \to \cdots \to x_{\ell-1} \to 1.$$

Halvings. The halving counts $k_1, \ldots, k_\ell$ of the cycle are defined by

$$qx_{i-1} + 1 = 2^{k_i} x_i \qquad (1 \leq i \leq \ell,\ x_\ell = x_0),$$

so $k_i = v_2(q \cdot x_{i-1} + 1)$. The total number of halvings is $h = k_1 + \cdots + k_\ell$, and $S_i = k_1 + \cdots + k_i$.

The δ window. Let $\ell \geq 2$ be an integer. On a cycle of length ℓ with total number of halvings h, the number δ is the fractional part of $2^{h/\ell}$. The integer q equals $\lfloor 2^{h/\ell} \rfloor$, so $2^{h/\ell} = q + \delta$ and $0 < \delta < 1$. The δ window is a lower bound and an upper bound on δ.

Near misses. For an element x_i other than 1 on a cycle of length ℓ with starting value 1, set $n_i = ih - \ell S_i$. The near miss of x_i is $2^{n_i/\ell}$. The relative error ρ_i is defined by

$$x_i = 2^{n_i/\ell}(1 + \rho_i).$$

The restricted irrationality exponent. Let θ be an irrational real number, and let S be a finite set of positive integers. A real number μ is a restricted irrationality exponent of θ for S when there exists an explicit constant real number $c > 0$ and there exists an explicit constant integer $j_0 \geq 0$ such that, for every positive integer $s \in S$, every integer p, and every integer $j \geq j_0$,

$$\left|\theta - \frac{p}{s \cdot 2^j}\right| > c \cdot 2^{-\mu j}.$$

The golden ratio. The golden ratio is $\varphi = (1 + \sqrt{5})/2$.

The order of 2. For an odd positive integer m, the order of 2 modulo m is the smallest positive integer d such that $2^d \equiv 1 \pmod{m}$.

Big O. Big O is the Bachmann–Landau symbol O. Let f and g be real-valued functions of a real variable t, with $g(t) \neq 0$ for every large t. Write $f(t) = O(g(t))$ when there exists a constant real number $C > 0$, independent of t, and there exists a constant real number t_0, such that $|f(t)| \leq C\,|g(t)|$ for every $t \geq t_0$. The expression $O(g(t))$ stands for a real number of absolute value at most $C\,|g(t)|$. An equation $A = B + O(g(t))$ means $A - B = O(g(t))$.

Vinogradov's symbols. Vinogradov's symbols are $\ll$ and $\gg$. Let f and g be real-valued functions of a real variable t. Write $f \ll g$ when $f = O(g)$. Write $f \gg g$ when $g \ll f$.

Hardy's symbol. Hardy's symbol is $\asymp$. Write $f \asymp g$ when $f \ll g$ and $g \ll f$.

Less than or similar to. The symbol $\lesssim$ means less than or similar to, and the symbol $\gtrsim$ means greater than or similar to. Let f and g be positive real numbers. Write $f \lesssim g$ when f is at most a constant real number times g, and that constant is not named. Write $f \gtrsim g$ when $g \lesssim f$.

3 The structure of a cycle of length 3

The three steps of length 3. On a cycle of length 3 with starting value 1, write x for x_1 and y for x_2. The cycle is $1 \to x \to y \to 1$. The three steps are $q + 1 = 2^{k_1}x$, $qx + 1 = 2^{k_2}y$, and $qy + 1 = 2^{k_3}$.

Lemma 1 (lower bound for k_3). $k_3 \geq k_1 + 3$.

Proof. The odd elements other than 1 are at least 3, so $x \geq 3$ and $y \geq 3$. The first step is $q + 1 = 2^{k_1}x$, and therefore $q + 1 \geq 3 \cdot 2^{k_1}$, that is $q \geq 3 \cdot 2^{k_1} - 1$. The third step is $2^{k_3} = qy + 1$, so

$$2^{k_3} \geq (3 \cdot 2^{k_1} - 1) \cdot 3 + 1 = 9 \cdot 2^{k_1} - 2.$$

Since $2^{k_1} \geq 2$, we have $9 \cdot 2^{k_1} - 2 \geq 8 \cdot 2^{k_1} = 2^{k_1+3}$, and hence $k_3 \geq k_1 + 3$. □

Lemma 2 (cycle equation).

$$2^h = q^3 + q^2 + 2^{k_1}q + 2^{k_1+k_2}.$$

Proof. Since $h = k_1 + k_2 + k_3$,

$$2^h = 2^{k_1+k_2} \cdot 2^{k_3}.$$

The third step is $qy + 1 = 2^{k_3}$, so

$$2^h = 2^{k_1+k_2}(qy + 1) = 2^{k_1}(q \cdot 2^{k_2}y + 2^{k_2}).$$

The second step is $qx + 1 = 2^{k_2}y$, and therefore

$$2^h = 2^{k_1}(q(qx + 1) + 2^{k_2}) = 2^{k_1}(q^2x + q + 2^{k_2}) = q^2(2^{k_1}x) + 2^{k_1}q + 2^{k_1+k_2}.$$

The first step is $q + 1 = 2^{k_1}x$, so

$$2^h = q^2(q + 1) + 2^{k_1}q + 2^{k_1+k_2} = q^3 + q^2 + 2^{k_1}q + 2^{k_1+k_2}.$$

□

Lemma 3 (between consecutive cubes). $q^3 < 2^h < (q+1)^3$. Hence 3 does not divide h and $q = \lfloor 2^{h/3} \rfloor$.

Proof. Lemma 2 gives $2^h = q^3 + q^2 + 2^{k_1}q + 2^{k_1+k_2}$. The terms q^2, $2^{k_1}q$, and $2^{k_1+k_2}$ are positive, so $2^h > q^3$.

Since $h = k_1 + k_2 + k_3$,

$$2^h = 2^{k_1} \cdot 2^{k_2} \cdot 2^{k_3}.$$

The three steps give $2^{k_1} = (q+1)/x$, $2^{k_2} = (qx+1)/y$, and $2^{k_3} = qy + 1$, so

$$2^h = \frac{q+1}{x} \cdot \frac{qx+1}{y} \cdot (qy+1) = (q+1) \cdot \frac{qx+1}{x} \cdot \frac{qy+1}{y}.$$

Since $(qx+1)/x = q + 1/x$ and $(qy+1)/y = q + 1/y$,

$$2^h = (q+1)\left(q + \frac{1}{x}\right)\left(q + \frac{1}{y}\right).$$

The elements x and y are distinct from the starting value 1, so $x > 1$ and $y > 1$. Hence $1/x < 1$ and $1/y < 1$, so $q + 1/x < q + 1$ and $q + 1/y < q + 1$, and $2^h < (q+1)^3$.

A power of 2 strictly between consecutive cubes is not a cube, so $3 \nmid h$. Since $q^3 < 2^h < (q+1)^3$, $q = \lfloor 2^{h/3} \rfloor$. □

Lemma 4 (2-adic valuation of $x-1$). Let t be the integer $v_2(x-1)$, and let s be the odd positive integer $(x-1)/2^t$. Then either $t = 2k_1$, or $t < k_1$ and $k_1 = t + v_2(s+1)$, or $t = k_1$.

Proof. The first step is $q + 1 = 2^{k_1}x$, so $q = 2^{k_1}x - 1$.

Substitute $q = 2^{k_1}x - 1$ into the second step $2^{k_2}y = qx + 1$:

$$2^{k_2}y = (2^{k_1}x - 1)x + 1 = 2^{k_1}x^2 - x + 1 = 2^{k_1}x^2 - (x-1).$$

Suppose $t > k_1$. Since x is odd, $v_2(2^{k_1}x^2) = k_1$, while $v_2(x-1) = t > k_1$, so

$$2^{k_1}x^2 - (x-1) = 2^{k_1}(x^2 - 2^{t-k_1}s),$$

and $x^2 - 2^{t-k_1}s$ is odd, since x^2 is odd and $2^{t-k_1}s$ is even. Thus the 2-adic valuation of $2^{k_2}y$ is k_1. Since y is odd, $k_2 = k_1$, so $2^{k_1}y = 2^{k_1}x^2 - (x-1)$.

Let $w = (x-1)/2^{k_1}$. Then w is even, since $t > k_1$, and $x = 1 + 2^{k_1}w$. Also $y = x^2 - w$, so

$$y = (1 + 2^{k_1}w)^2 - w = 1 + w(2^{k_1+1} - 1) + 2^{2k_1}w^2.$$

The first step gives $q = 2^{k_1}x - 1 = 2^{k_1} + 2^{2k_1}w - 1$. Multiplying these expressions for q and y and adding 1 gives

$$qy + 1 = 2^{k_1} + w(1 - 3 \cdot 2^{k_1}) + 2^{2k_1}w(3 - 2w) + 3 \cdot 2^{3k_1}w^2 + 2^{4k_1}w^3.$$

The terms $2^{2k_1}w(3-2w)$, $3\cdot 2^{3k_1}w^2$, and $2^{4k_1}w^3$ are divisible by $2^{2k_1+v_2(w)}$, and $1-3\cdot 2^{k_1}$ is odd, so $v_2(w(1-3\cdot 2^{k_1})) = v_2(w)$. If $v_2(w) \neq k_1$, then $v_2(qy+1) = \min(k_1, v_2(w)) \leq k_1$, contradicting Lemma 1. Hence $v_2(w) = k_1$, that is $t = 2k_1$.

Suppose $t < k_1$. Then $2^{k_2}y = 2^{k_1}x^2 - 2^t s = 2^t(2^{k_1-t}x^2 - s)$. The factor $2^{k_1-t}x^2 - s$ is odd, since $2^{k_1-t}x^2$ is even and s is odd, so $k_2 = t$ and $y = 2^{k_1-t}x^2 - s$. Since $x = 1 + 2^t s$, we have $q = 2^{k_1}x - 1 = (2^{k_1}-1) + 2^{k_1+t}s$, and therefore

$$qy + 1 = (2^{k_1}-1)y + 2^{k_1+t}sy + 1.$$

Substitute $y = 2^{k_1-t}x^2 - s$:

$$qy + 1 = (2^{k_1}-1)\,2^{k_1-t}x^2 + (1+s) - 2^{k_1}s + 2^{k_1+t}sy.$$

Now $2^{k_1}s$ and $2^{k_1+t}sy$ are divisible by 2^{k_1}, and $v_2((2^{k_1}-1)\,2^{k_1-t}x^2) = k_1 - t$, since $2^{k_1}-1$ and x are odd. If $v_2(s+1) \neq k_1 - t$, then $v_2(qy+1) < k_1$, contradicting Lemma 1. Hence $v_2(s+1) = k_1 - t$, that is $k_1 = t + v_2(s+1)$.

If $t = k_1$, that is the remaining case. □

Lemma 5 (lower bound for x). $2^{k_1} \leq 2(x-1)$ and $x > \sqrt{q/2}$.

Proof. If $t = 2k_1$, then $2^{2k_1} = 2^t \leq x-1$, so $2^{k_1} \leq x-1 \leq 2(x-1)$. If $t < k_1$ and $k_1 = t + v_2(s+1)$, then $2^{k_1} = 2^t \cdot 2^{v_2(s+1)} \leq 2^t(s+1) = (x-1)+2^t \leq 2(x-1)$. If $t = k_1$, then $2^{k_1} = 2^t \leq x-1 \leq 2(x-1)$. In every case $2^{k_1} \leq 2(x-1)$. The first step is $q+1 = 2^{k_1}x$, so

$$q + 1 \leq 2x(x-1) = 2x^2 - 2x.$$

Hence $q < 2x^2$, so $x > \sqrt{q/2}$. □

Lemma 6. Let $\ell \geq 2$. Let i be an integer with $1 \leq i \leq \ell - 1$. On a cycle of length ℓ with starting value 1,

$$2^{k_{i+1}}x_{i-1}\,x_{i+1} - 2^{k_i}x_i^2 = x_{i-1} - x_i.$$

Also $|x_{i-1} - x_i| \geq 2^{\min(k_i, k_{i+1})}$. If $k_i \leq k_{i+1}$, then $x_i > q$ when $x_i < x_{i-1}$, and $x_i^2 > q\,x_{i-1}$ when $x_i > x_{i-1}$.

Proof. Since $qx_i + 1 = 2^{k_{i+1}}x_{i+1}$, we have $q = (2^{k_{i+1}}x_{i+1} - 1)/x_i$. Substitute $q = (2^{k_{i+1}}x_{i+1} - 1)/x_i$ into $qx_{i-1} + 1 = 2^{k_i}x_i$ and multiply by x_i:

$$2^{k_{i+1}}x_{i-1}\,x_{i+1} - x_{i-1} + x_i = 2^{k_i}x_i^2,$$

and therefore $2^{k_{i+1}}x_{i-1}\,x_{i+1} - 2^{k_i}x_i^2 = x_{i-1} - x_i$.

The integers x_{i-1}, x_i, and x_{i+1} are odd, so $x_{i-1}x_{i+1}$ and x_i^2 are odd. The 2-adic valuation of $2^{k_{i+1}}x_{i-1}x_{i+1}$ is k_{i+1}, and the 2-adic valuation of $2^{k_i}x_i^2$ is k_i. If $k_i \neq k_{i+1}$, the 2-adic valuation of $2^{k_{i+1}}x_{i-1}x_{i+1} - 2^{k_i}x_i^2$ is $\min(k_i, k_{i+1})$. If $k_i = k_{i+1}$, then $x_{i-1}x_{i+1} - x_i^2$ is even, since it is a difference of two odd integers, so the 2-adic valuation of $2^{k_i}(x_{i-1}x_{i+1} - x_i^2)$ is at least k_i. In

either case $2^{\min(k_i,k_{i+1})}$ divides $x_{i-1}-x_i$. The elements of the cycle are distinct, so $x_{i-1}\neq x_i$ and $|x_{i-1}-x_i|\geq 2^{\min(k_i,k_{i+1})}$.

Suppose $k_i\leq k_{i+1}$. Then $|x_{i-1}-x_i|\geq 2^{k_i}=(qx_{i-1}+1)/x_i$. If $x_i>x_{i-1}$, then $x_i(x_i-x_{i-1})\geq qx_{i-1}+1$. Since $x_{i-1}\geq 1$, we have $x_i^2>x_i(x_i-x_{i-1})$, and therefore $x_i^2>qx_{i-1}$. If $x_i<x_{i-1}$ and $x_i\leq q$, then $x_{i-1}x_i\leq qx_{i-1}$, so

$$x_i(x_{i-1}-x_i)=x_{i-1}x_i-x_i^2\leq qx_{i-1}-x_i^2<qx_{i-1}+1.$$

This contradicts $x_i(x_{i-1}-x_i)\geq qx_{i-1}+1$. Hence $x_i>q$. □

Lemma 7. *$y>x$ and $y^2>qx$. In particular $2^{k_2}<q+1$, so $2^{k_1+k_2}\leq 2^{k_1}q$.*

Proof. Apply Lemma 6 with $i=2$. Then $x_1=x$, $x_2=y$, and $x_3=1$. Suppose $k_3<k_2$. Lemma 6 gives $|x-y|\geq 2^{k_3}$. The third step is $qy+1=2^{k_3}$, so $|x-y|\geq qy+1$. If $y>x$, then $y-x\geq qy+1$, and $y(1-q)\geq x+1>0$. Here $q\geq 3$ and $y>0$, so $y(1-q)<0$, a contradiction. Thus $x>y$, and $x-y\geq qy+1$, so $x\geq(q+1)y+1>qy$. The element y is an odd integer at least 3, so $x>3q$. The first step is $q+1=2^{k_1}x$ with $k_1\geq 1$, so $x\leq(q+1)/2$. Since $q\geq 3$, we have $3q>(q+1)/2$, and therefore $x>(q+1)/2\geq x$, a contradiction. Thus $k_2\leq k_3$.

If $y<x$, Lemma 6 gives $y>q$. Also $x=(q+1)/2^{k_1}\leq(q+1)/2<q$, so $y>q>x$, a contradiction. The elements of the cycle are distinct, so $y\neq x$. Hence $y>x$, and Lemma 6 gives $y^2>qx$. Since $y>x$,

$$2^{k_2}=\frac{qx+1}{y}<\frac{qx+1}{x}=q+\frac{1}{x}<q+1.$$

The integer 2^{k_2} is at most q, so $2^{k_1+k_2}\leq 2^{k_1}q$. □

4 The δ window and the near miss for x

Lemma 8 (the δ window). Let $N=27\cdot 2^h-(3q+1)^3$. Then

$$N=9q(3\cdot 2^{k_1}-1)+27\cdot 2^{k_1+k_2}-1,\qquad 0<N<54\cdot 2^{k_1}q,\qquad 0<\delta-\frac{1}{3}\leq\frac{N}{81q^2}<\frac{2^{k_1+1}}{3q}.$$

Proof. Lemma 2 gives $2^h=q^3+q^2+2^{k_1}q+2^{k_1+k_2}$. Expanding $(3q+1)^3=27q^3+27q^2+9q+1$ and subtracting from $27\cdot 2^h$ gives

$$N=9q(3\cdot 2^{k_1}-1)+27\cdot 2^{k_1+k_2}-1.$$

Since $2^{k_1}\geq 2$, the formula gives $N\geq 9q(3\cdot 2-1)-1=45q-1>0$. Lemma 7 gives $2^{k_1+k_2}\leq 2^{k_1}q$, so

$$N<9q(3\cdot 2^{k_1})+27\cdot 2^{k_1}q=54\cdot 2^{k_1}q.$$

Since $2^{h/3}=q+\delta$ and $(3q+1)^3=27(q+1/3)^3$,

$$N=27\big[(q+\delta)^3-(q+1/3)^3\big].$$

The factor $(q+\delta)^3-(q+1/3)^3$ equals $(\delta-1/3)$ times $(q+\delta)^2+(q+\delta)(q+1/3)+(q+1/3)^2$. Each of those three terms is greater than q^2, so their sum is greater than $3q^2$. Thus $N>0$ gives $\delta>1/3$, and $N\geq 81q^2(\delta-1/3)$, so $\delta-1/3\leq N/(81q^2)$. Also

$$\frac{N}{81q^2}<\frac{54\cdot 2^{k_1}q}{81q^2}=\frac{2^{k_1+1}}{3q}.$$

□

Lemma 9 (near miss for x). Let $f=h-3k_1$. Then $3\nmid f$, and

$$\left|x^3-2^f\right|\leq\frac{3.2\,x^2}{2^{k_1}},\qquad \left|x-2^{f/3}\right|\leq\frac{1.6}{2^{k_1}}.$$

Proof. Since $q+1=2^{k_1}x$, we have $2^{3k_1}x^3=(q+1)^3$.

Let $E=2q^2+(3-2^{k_1})q+1-2^{k_1+k_2}$. Lemma 2 gives $2^h=q^3+q^2+2^{k_1}q+2^{k_1+k_2}$, so

$$2^{3k_1}x^3-2^h=(q+1)^3-2^h=2q^2+(3-2^{k_1})q+1-2^{k_1+k_2}=E.$$

Since $2^{k_1}\geq 2$, either $2^{k_1}=2$ and $|3-2|=1$, or $2^{k_1}\geq 4$ and $|3-2^{k_1}|=2^{k_1}-3$. In both cases $|3-2^{k_1}|\leq 2^{k_1}$. Since $q<2^{k_1}x$ and, by Lemma 7, $2^{k_1+k_2}\leq 2^{k_1}q<2^{2k_1}x$,

$$|E|<2\cdot 2^{2k_1}x^2+2\cdot 2^{2k_1}x+1=2^{2k_1}x^2\left(2+\frac{2}{x}+\frac{1}{2^{2k_1}x^2}\right).$$

Since $x\geq 3$ and $2^{k_1}\geq 2$,

$$2+\frac{2}{x}+\frac{1}{2^{2k_1}x^2}\leq 2+\frac{2}{3}+\frac{1}{36}=\frac{97}{36}<3.2,$$

so $|E|<3.2\cdot 2^{2k_1}x^2$. Dividing by 2^{3k_1} gives $\left|x^3-2^f\right|<3.2\,x^2/2^{k_1}$, and in particular

$$\left|x^3-2^f\right|\leq\frac{3.2\,x^2}{2^{k_1}}.$$

Since $f=h-3k_1$, we have $f\equiv h\pmod 3$. Lemma 3 says that 3 does not divide h, so 3 does not divide f.

$$\left|x^3-2^f\right|=\left|x-2^{f/3}\right|(x^2+x\cdot 2^{f/3}+2^{2f/3}).$$

Suppose $\left|x-2^{f/3}\right|\geq 1$. For every real number z, $\left|x^3-z^3\right|$ increases as z moves away from x, so

$$\left|x^3-2^f\right|\geq\min(x^3-(x-1)^3,\ (x+1)^3-x^3)=3x^2-3x+1,$$

because $(x+1)^3-x^3=3x^2+3x+1>3x^2-3x+1$. Since $x\geq 3$, we have $3x^2-3x+1\geq 2x^2$. Since $2^{k_1}\geq 2$, we have $3.2\,x^2/2^{k_1}\leq 1.6x^2<2x^2$. So $\left|x^3-2^f\right|\geq 2x^2$ and $\left|x^3-2^f\right|<2x^2$, which is impossible.

Hence $\left|x-2^{f/3}\right|<1$, so $2^{f/3}>x-1$. Since $x\geq 3$, we have $x-1\geq 2x/3$, and therefore $2^{f/3}>2x/3$. Then

$$x^2+x\cdot 2^{f/3}+2^{2f/3}>x^2+\frac{2x^2}{3}+\frac{4x^2}{9}=\frac{19}{9}x^2>2x^2,$$

so

$$\left|x - 2^{f/3}\right| = \frac{\left|x^3 - 2^f\right|}{x^2 + x \cdot 2^{f/3} + 2^{2f/3}} < \frac{3.2\, x^2}{2^{k_1} \cdot 2x^2} = \frac{1.6}{2^{k_1}}.$$

In particular $\left|x - 2^{f/3}\right| \le 1.6/2^{k_1}$. □

5 Proof of Theorem 1

We use the following inequality, obtained in the proof of Theorem 2.1 of Bauer and Bennett (2018) from their inequality (5.3), Lemma 4.1 and Proposition 5.1, before they round the exponent to $2^{-1.62k}$ in (2.1).

Theorem (Bauer and Bennett (2018)). Let $j \ge 11000$, $r \in \{1, 2\}$, $s \in \{1, 3\}$ and $n = 3j + r$. For every integer p,

$$\left|2^{r/3} - \frac{p}{s\, 2^j}\right| \ge \left|1 - \frac{p}{s\, 2^{n/3}}\right| > 2^{-5} \cdot 16166467234^{-(n+77)/63}.$$

The factor 2^{-5} is a weakening of the contour bound behind their inequality (5.3). The denominator of (5.3) contributes $1/(1.26 \cdot 2^{14/3}) = 2^{-5.00009\ldots}$, and the contour estimate for $|P_{n_1,\delta}|$ has constant $1.2528\ldots$ in place of 1.26, so that $1.2528\ldots \cdot 2^{14/3}$ has base-2 logarithm $4.9918 < 5$. The formula they display omits the factor, though it is absorbed in their final statement. Per unit of j the exponent is $\mu = 3\log_2(16166467234)/63 \approx 1.61487$.

Proof of Theorem 1. Suppose $q \ge 2^{22001}$. Write $a = k_1$, $h = 3m + r$ and $f = h - 3a = 3j' + r'$, where $r, r' \in \{1, 2\}$ by Lemmas 3 and 9. Lemma 5 gives $x > \sqrt{q/2} \ge 2^{11000}$. Since q is odd, $a \ge 1$, and Lemma 9 gives $|x - 2^{f/3}| \le 0.8$. Also $2^{f/3} \le 2^{j'+2/3}$. If $j' \le 10999$, then $x \le 2^{10999+2/3} + 0.8$. Now $(8/5)^3 = 512/125 > 4$, so $2^{2/3} < 8/5$, and $2^{11000} - (8/5)\, 2^{10999} = (2/5)\, 2^{10999} > 0.8$. Thus $x < 2^{11000}$, a contradiction, and $j' \ge 11000$. Likewise $(5/4)^3 = 125/64 < 2$, so $2^{1/3} > 5/4$ and $2^{j'}/4 > 0.8$, which gives $x \ge 2^{j'+1/3} - 0.8 > 2^{j'} + 2^{j'}/4 - 0.8 > 2^{j'}$. Hence $x \ge 2^{j'} + 1$ and $q = 2^a x - 1 \ge 2^{a+j'} + 2^a - 1 \ge 2^{a+j'} + 1$. From Lemma 3, $2^h > q^3$, so $2^{h/3} > q > 2^{a+j'}$ and $m + r/3 > a + j'$. As $r \le 2$, this forces $m \ge a + j' \ge 11000$.

1. **The δ window.** Apply Theorem (Bauer and Bennett (2018)) with $s = 3$, $p = 3q + 1$, $j = m$. Since $2^{h/3} = q + \delta$,

   $$1 - \frac{3q+1}{3 \cdot 2^{h/3}} = \frac{3(q+\delta) - (3q+1)}{3 \cdot 2^{h/3}} = \frac{\delta - 1/3}{2^{h/3}},$$

   so $\delta - 1/3 > 2^m \cdot 2^{-5} \cdot 16166467234^{-(h+77)/63}$. With Lemma 8 and $q \ge 2^m$,

   $$2^a > 1.5 \cdot 2^{2m-5} \cdot 16166467234^{-(h+77)/63}.$$

2. **The near miss for x.** Apply Theorem (Bauer and Bennett (2018)) with $s = 1$, $p = x$, $j = j'$. By Lemma 9, $|1 - x/2^{f/3}| \le 1.6/(2^a \cdot 2^{f/3})$, so

   $$2^a < 1.6 \cdot 2^{5-f/3} \cdot 16166467234^{(f+77)/63}.$$

3. **Combining.** With $h = f + 3a$, $m \geq a + j'$ and $j' \geq (f-2)/3$, the δ window gives $a > 0.20878\,f - 76.8$ and the near miss for x gives $a < 0.20496\,f + 47.2$. These are incompatible once $f \geq 32{,}461$, that is $j' \geq 10{,}820$.

So any cycle of length 3 with starting value 1 has $q < 2^{22001}$. Lemma 3 gives $2^h < (q+1)^3$. Since q is an integer, $q + 1 \leq 2^{22001}$, so $2^h < 2^{66{,}003}$ and $h < 66{,}003$. Each such h gives a single candidate $q = \lfloor 2^{h/3} \rfloor$. Appendix E tests every $h \leq 200{,}000$ and finds no cycle, and the digit scan there rules out every such cycle with $q < 2^{1{,}071{,}999{,}953}$. □

Why the golden ratio appears. The proof squeezes x between two bounds, both governed by the exponent μ in Theorem (Bauer and Bennett (2018)).

- **The δ window keeps x small.** By Lemma 8, $2^{h/3}$ exceeds $q + 1/3$ by about $1/(3x)$, so $(3q+1)/(3 \cdot 2^m)$ approximates $2^{r/3}$ with error about $1/(xq)$. Theorem (Bauer and Bennett (2018)) forbids errors below about $q^{-\mu}$, so $x \lesssim q^{\mu-1}$.

- **The near miss for x keeps x large.** By Lemma 9, $x/2^{j'}$ approximates $2^{r'/3}$ with error about $1/q$, where $2^{j'} \asymp x$. Theorem (Bauer and Bennett (2018)) forbids errors below about $x^{-\mu}$, so $x \gtrsim q^{1/\mu}$.

Both bounds can hold only if $1/\mu \leq \mu - 1$, that is $\mu^2 - \mu - 1 \geq 0$, whose positive root is $\varphi = (1+\sqrt{5})/2$. With $\mu = 1.61487$ the bounds read $q^{0.6192} \lesssim x \lesssim q^{0.6149}$, which are incompatible once q is large; the explicit constants in the δ window, the near miss for x, and the comparison of those two bounds make this precise for $q \geq 2^{22001}$. The two bounds are tied together by $q + 1 = 2^{k_1} x$. If x is large, then 2^{k_1} is small, and the error in $(3q+1)/(3 \cdot 2^m)$ as an approximation to $2^{r/3}$ falls below the bound forbidden by Theorem (Bauer and Bennett (2018)). If x is small, then 2^{k_1} is large, and the error in $x/2^{j'}$ as an approximation to $2^{r'/3}$ falls below that same bound.

The threshold φ comes from this argument; the exponent μ comes from Bauer and Bennett (2018), who needed only an exponent below 2 for their own purposes. Their stated value 1.62 lies just above φ and would leave room for x; the unrounded 1.61487 lies just below it.

6 Longer cycles

Lemma 10 (between consecutive ℓ-th powers). For a cycle of length ℓ with starting value 1, let $\sigma = \sum_{i=1}^{\ell-1} 1/x_i$. Then

$$2^h = (q+1) \prod_{i=1}^{\ell-1} \left(q + \frac{1}{x_i} \right).$$

If $\ell \geq 2$, then $q^\ell < 2^h < (q+1)^\ell$, ℓ does not divide h, and $q = \lfloor 2^{h/\ell} \rfloor$. Also $\delta - 1/\ell = \sigma/\ell + O(1/q)$.

Proof. For $1 \leq i \leq \ell$, $2^{k_i} x_i = q x_{i-1} + 1$. Since $h = k_1 + \cdots + k_\ell$, $x_0 = 1$ and $x_\ell = 1$,

$$2^h = \prod_{i=1}^{\ell} \frac{q x_{i-1} + 1}{x_i} = (q+1) \prod_{i=1}^{\ell-1} \left(q + \frac{1}{x_i} \right).$$

When $\ell = 1$ there is no index i with $1 \le i \le \ell - 1$, so that product has no factors and equals 1. Thus $2^h = q + 1$.

Suppose $\ell \ge 2$. Each x_i with $1 \le i \le \ell - 1$ is an odd integer at least 3, so $q < q + 1/x_i < q + 1$. Therefore

$$q^\ell < (q+1)\, q^{\ell-1} < (q+1) \prod_{i=1}^{\ell-1} \left(q + \frac{1}{x_i} \right) < (q+1)^\ell.$$

The product $(q+1)\prod_{i=1}^{\ell-1}(q + 1/x_i)$ equals 2^h, so $q^\ell < 2^h < (q+1)^\ell$. A power of 2 strictly between consecutive ℓ-th powers is not an ℓ-th power, so ℓ does not divide h, and $q = \lfloor 2^{h/\ell} \rfloor$.

Since $2^{h/\ell} = q + \delta$, we have $(q+\delta)^\ell = 2^h$. Dividing by q^ℓ gives

$$\left(1 + \frac{\delta}{q}\right)^\ell = \left(1 + \frac{1}{q}\right) \prod_{i=1}^{\ell-1} \left(1 + \frac{1}{q x_i}\right).$$

Expanding the product, $\prod_{i=1}^{\ell-1}(1 + 1/(qx_i)) = 1 + \sigma/q + O(1/q^2)$: the sum of the factors $1/(qx_i)$ is σ/q, and every product of two or more such factors is $O(1/q^2)$, with the constant depending only on ℓ. Thus

$$\left(1 + \frac{1}{q}\right)\left(1 + \frac{\sigma}{q} + O(1/q^2)\right) = 1 + \frac{1+\sigma}{q} + O(1/q^2).$$

Since $0 < \delta < 1$, the binomial expansion gives $(1 + \delta/q)^\ell = 1 + \ell\delta/q + O(1/q^2)$. From $1 + \ell\delta/q + O(1/q^2) = 1 + (1+\sigma)/q + O(1/q^2)$ we have $\ell\delta/q = (1+\sigma)/q + O(1/q^2)$, so $\delta - 1/\ell = \sigma/\ell + O(1/q)$. □

Lemma 11 (the relative error ρ_i). For $1 \le i \le \ell - 1$, let

$$\varepsilon_i = \frac{1}{q} + \sum_{u=1}^{i-1} \frac{1 + \varepsilon_u}{q x_u}.$$

Then $x_i\, 2^{S_i} = q^i(1 + \varepsilon_i)$. Since $n_i = ih - \ell S_i$, we have $x_i = 2^{n_i/\ell}(1 + \rho_i)$. For $q \ge \ell$,

$$-\frac{i}{q+1} < \rho_i \le \frac{2i+1}{3q}.$$

In particular $|\rho_i| \le (\ell - 1)/q$. If $q \ge \ell$ and ℓ divides n_i, then $x_i \ge q/(\ell - 1) - 1$.

Proof. For $i = 1$, we have $\varepsilon_1 = 1/q$ and $x_1 2^{k_1} = q + 1 = q(1 + 1/q)$.

Suppose $i \ge 2$ and $x_{i-1} 2^{S_{i-1}} = q^{i-1}(1 + \varepsilon_{i-1})$. Since $2^{k_i} x_i = q x_{i-1} + 1$ and $S_i = S_{i-1} + k_i$,

$$x_i 2^{S_i} = 2^{S_{i-1}}(q x_{i-1} + 1) = q^i(1 + \varepsilon_{i-1}) + 2^{S_{i-1}}.$$

The inductive equation gives $2^{S_{i-1}}/q^i = (1 + \varepsilon_{i-1})/(q x_{i-1})$. The definition of ε_i gives $\varepsilon_i = \varepsilon_{i-1} + (1 + \varepsilon_{i-1})/(q x_{i-1})$. Therefore $x_i 2^{S_i} = q^i(1 + \varepsilon_i)$.

Since $n_i = ih - \ell S_i$ and $x_i = 2^{n_i/\ell}(1 + \rho_i)$,

$$1 + \rho_i = \frac{x_i 2^{S_i}}{2^{ih/\ell}} = \left(\frac{q}{q+\delta}\right)^i (1 + \varepsilon_i),$$

where $2^{h/\ell} = q + \delta$. For $\ell \geq 2$, Lemma 10 gives $0 < \delta < 1$.

Now $\varepsilon_1 = 1/q = (2 \cdot 1 + 1)/(3q)$. If $\varepsilon_u \leq 1$ for every $u < i$, then $(1 + \varepsilon_u)/(qx_u) \leq 2/(3q)$, because $x_u \geq 3$, and

$$\varepsilon_i \leq \frac{1}{q} + \frac{2}{q} \cdot \frac{i-1}{3} = \frac{2i+1}{3q}.$$

For $q \geq \ell$ and $i \leq \ell - 1$, we have $(2i+1)/(3q) \leq (2\ell-1)/(3\ell) < 1$, so $\varepsilon_i \leq 1$. Thus $\varepsilon_i \leq (2i+1)/(3q)$ for every $i \leq \ell - 1$. Since $0 < \delta < 1$, we have $\big(q/(q+1)\big)^i < \big(q/(q+\delta)\big)^i$, and $\big(q/(q+1)\big)^i = \big(1 - 1/(q+1)\big)^i \geq 1 - i/(q+1)$. Also $\big(q/(q+\delta)\big)^i \leq 1$ and $1 + \varepsilon_i \leq 1 + (2i+1)/(3q)$, so

$$1 - \frac{i}{q+1} \leq \left(1 - \frac{1}{q+1}\right)^i < \left(\frac{q}{q+\delta}\right)^i (1 + \varepsilon_i) \leq 1 + \frac{2i+1}{3q}.$$

Here $\big(q/(q+\delta)\big)^i(1+\varepsilon_i) = 1 + \rho_i$, so $-i/(q+1) < \rho_i \leq (2i+1)/(3q)$. Also $(2i+1)/(3q) \leq (\ell-1)/q$ and $i/(q+1) < (\ell-1)/q$, so $|\rho_i| \leq (\ell-1)/q$.

Suppose $q \geq \ell$ and ℓ divides n_i, so n_i/ℓ is an integer. If $n_i \geq 0$, then $2^{n_i/\ell}$ is a positive integer power of 2, hence 1 or even, while x_i is odd and at least 3, so $|x_i - 2^{n_i/\ell}| \geq 1$. If $n_i < 0$, then $2^{n_i/\ell} \leq 1/2$ and $x_i \geq 3$, so $|x_i - 2^{n_i/\ell}| \geq 5/2$. In either case $|\rho_i| = |x_i - 2^{n_i/\ell}| \cdot 2^{-n_i/\ell} \geq 2^{-n_i/\ell}$. Since $2^{n_i/\ell} = x_i/(1 + \rho_i)$ and $|\rho_i| \leq (\ell-1)/q < 1$,

$$2^{-n_i/\ell} = \frac{1 + \rho_i}{x_i} \geq \frac{1 - |\rho_i|}{x_i}.$$

Hence $|\rho_i| \geq (1 - |\rho_i|)/x_i$, so $x_i \geq 1/|\rho_i| - 1 \geq q/(\ell-1) - 1$. □

The outline in the introduction is the case $R = \{1, \dots, \ell - 1\}$.

Theorem 2. Let $\ell \geq 2$, let $R \subseteq \{1, \dots, \ell - 1\}$, and let $\varphi = (1 + \sqrt{5})/2$. Suppose that $0 < \mu < \varphi$ and that, for every $r \in R$, μ is a restricted irrationality exponent of $2^{r/\ell}$ for the set $\{1, \ell\}$, with the same explicit constant real number $c > 0$ and the same explicit constant integer $j_0 \geq 0$. Then there is a number Q, computable from ℓ, μ, c and j_0, with the following property. If $q > Q$ is odd and $1 = x_0 \to x_1 \to \cdots \to x_{\ell-1} \to 1$ is a cycle of C of length ℓ, with $q \cdot x_{i-1} + 1 = 2^{k_i} \cdot x_i$ for $1 \leq i \leq \ell$ (where $x_\ell = 1$), then some multiple of $h = k_1 + \cdots + k_\ell$ has a nonzero residue modulo ℓ outside R.

Proof of Theorem 2. Suppose that every multiple of h has its nonzero residues modulo ℓ in R. Write $h = \ell m + r$ with $r \in R$, and assume $q \geq 2(\ell - 1)$, so Lemma 11 gives $|\rho_i| \leq 1/2$. The δ window uses $2^{r/\ell}$, and x_i uses $2^{n_i/\ell}$ with $n_i \equiv ir \pmod{\ell}$, so the residue modulo ℓ of each of these exponents lies in R. Set

$$A = 2^{1/\ell - 1}\Big(\frac{c}{2(\ell-1)}\Big)^{1/\mu}, \qquad \gamma = 1 - \mu + \frac{1}{\mu} > 0,$$

the last because $\mu^2 - \mu - 1 < 0$.

If $q < 2^{j_0+1}$, stop: this is already a bound. If $q \geq 2^{j_0+1}$, then $2^m = (q + \delta)/2^{r/\ell} > q/2 \geq 2^{j_0}$, so $m \geq j_0$ and the hypothesis applies to the δ window.

$$\Big|\delta - \frac{1}{\ell}\Big| = 2^m \Big|2^{r/\ell} - \frac{\ell q + 1}{\ell \cdot 2^m}\Big| > c \cdot 2^{(1-\mu)m}.$$

If $\mu > 1$ then $2^m < q+1$ gives $|\delta - 1/\ell| > c(q+1)^{1-\mu} \geq c\,2^{1-\mu}q^{1-\mu}$. If $\mu \leq 1$ then $2^m > q/2$ gives $|\delta - 1/\ell| > c(q/2)^{1-\mu} \geq c/2$.

For each x_i with $\ell \nmid n_i$, write $n_i = \ell j + r'$ and $x_i/2^j = 2^{r'/\ell}(1+\rho_i)$, so $|2^{r'/\ell} - x_i/2^j| = 2^{r'/\ell}|\rho_i|$. Also $x_i/3 \leq 2^j \leq 2x_i$. If $j < j_0$, then $x_i < 3 \cdot 2^{j_0}$. The finitely many triples (r', j, x) with $r' \in R$, $0 \leq j < j_0$ and x odd, $3 \leq x < 3 \cdot 2^{j_0}$, each have a positive $|\rho| = |x \cdot 2^{-j-r'/\ell} - 1|$, and Lemma 11 forces $q \leq (\ell-1)/|\rho|$. Let Q_0 be the maximum of those finitely many values. If $j \geq j_0$, the hypothesis and $|\rho_i| \leq (\ell-1)/q$ give $c \cdot 2^{-\mu j} < 2(\ell-1)/q$, so

$$2^j > \Big(\frac{cq}{2(\ell-1)}\Big)^{1/\mu}, \qquad x_i \geq 2^{1/\ell - 1}\,2^j \geq A\,q^{1/\mu}.$$

If $\ell \mid n_i$, then $x_i \geq q/(\ell-1) - 1 \geq q/(2(\ell-1))$. Once $q > Q_0$, every x_i is at least $\min(Aq^{1/\mu},\ q/(2(\ell-1)))$, so

$$\sigma \leq (\ell-1)\,A^{-1}q^{-1/\mu} + \frac{2(\ell-1)^2}{q}.$$

Expanding the product in Lemma 10,

$$\Big(1+\frac{\delta}{q}\Big)^{\ell} = \Big(1+\frac{1}{q}\Big)\prod_{i=1}^{\ell-1}\Big(1+\frac{1}{qx_i}\Big).$$

The product is $1 + \sigma/q + \rho$ with $0 < \rho \leq (\sigma/q)^2 e^{\sigma/q}$. Comparing with the binomial expansion of the left side, and using $\sigma \leq (\ell-1)/3$ and $q \geq \ell$,

$$\Big|\delta - \frac{1}{\ell} - \frac{\sigma}{\ell}\Big| \leq \frac{\sigma}{\ell q} + \frac{\rho(q+1)}{\ell} + \frac{\ell\, e^{\ell/q}}{q} \leq \frac{4\ell}{q}.$$

Hence

$$|\delta - \frac{1}{\ell}| \leq A^{-1}q^{-1/\mu} + \left(\frac{2(\ell-1)^2}{\ell} + 4\ell\right)\frac{1}{q}.$$

If $\mu > 1$, then $1/q \leq q^{-1/\mu}$ for every integer $q \geq 1$, so

$$|\delta - \frac{1}{\ell}| \leq \left(A^{-1} + \frac{2(\ell-1)^2}{\ell} + 4\ell\right)q^{-1/\mu}.$$

Let Q_* be the positive real number

$$Q_* = \left(\frac{A^{-1} + 2(\ell-1)^2/\ell + 4\ell}{c\,2^{1-\mu}}\right)^{1/\gamma}.$$

Take $Q = \max(2(\ell-1),\ 2^{j_0+1},\ Q_0,\ Q_*)$. If $q > Q$, then $|\delta - 1/\ell| > c\,2^{1-\mu}q^{1-\mu}$ and $|\delta - 1/\ell| \leq (A^{-1} + 2(\ell-1)^2/\ell + 4\ell)q^{-1/\mu}$ cannot both hold.

If $\mu \leq 1$, then $q^{-1/\mu} \leq 1/q$ for every integer $q \geq 1$, so

$$|\delta - \frac{1}{\ell}| \leq \left(A^{-1} + \frac{2(\ell-1)^2}{\ell} + 4\ell\right)\frac{1}{q}.$$

Let Q_* be the positive real number

$$Q_* = \frac{2}{c}\left(A^{-1} + \frac{2(\ell-1)^2}{\ell} + 4\ell\right).$$

Take $Q = \max(2(\ell-1),\ 2^{j_0+1},\ Q_0,\ Q_*)$. If $q > Q$, then $\left|\delta - 1/\ell\right| > c/2$ and $\left|\delta - 1/\ell\right| \le (A^{-1} + 2(\ell-1)^2/\ell + 4\ell)/q$ cannot both hold. □

If the lower bound is a power of the whole denominator, the multiplier contributes only a constant real number. Suppose there exists an explicit constant real number $c_0 > 0$ and there exists an explicit constant integer $j_0 \ge 0$ such that, for every integer r with $1 \le r \le \ell - 1$, every integer p, every $s \in \{1, \ell\}$, and every integer $j \ge j_0$,

$$\left|2^{r/\ell} - \frac{p}{s \cdot 2^j}\right| > c_0 (s \cdot 2^j)^{-\mu}.$$

Since $\mu > 0$ and $1 \le s \le \ell$, we have $s^{-\mu} \ge \ell^{-\mu}$. The displayed inequality therefore yields the hypothesis of Theorem 2 with the explicit constant real number $c = c_0 \ell^{-\mu}$. The exponent of 2 remains μ. Ridout's theorem gives a lower bound of this form with $\mu = 1 + \varepsilon$ and with the power on the whole denominator $s \cdot 2^j$. It gives no explicit c_0 or j_0.

For $R = \{1, \dots, \ell - 1\}$ the conclusion is that no $q > Q$ has a cycle of length ℓ with starting value 1. For $\ell = 4$ and $R = \{2\}$, the multiples of h avoid the residues 1 and 3 exactly when h is even, so the conclusion is that h is odd.

For $\ell = 3$, Theorem (Bauer and Bennett (2018)) supplies the hypothesis of Theorem 2 with $R = \{1, 2\}$, $\mu \approx 1.61487$ and $j_0 = 11000$. So Theorem 1 is the case $\ell = 3$ of Theorem 2; Section 5 carries out this case with explicit constants.

Corollary 12. No cycle of length 4 with starting value 1 has total number of halvings $\equiv 2 \pmod 4$.

Proof. Write $h = 4m + 2$ for an integer $m \ge 0$. Then $n_1 \equiv n_3 \equiv 2 \pmod 4$ and $n_2 \equiv 0 \pmod 4$. For $i \in \{1, 3\}$, let j be the integer $(n_i - 2)/4$, so $x_i = \sqrt{2}\, 2^j (1 + \rho_i)$. Also $2^m \sqrt{2} = q + \delta$, and

$$\left|\delta - \frac{1}{4}\right| = 2^m \left|\sqrt{2} - \frac{4q+1}{2^{m+2}}\right|.$$

Bauer and Bennett's Corollary 1.6 (2002) gives $|\sqrt{2} - p/2^j| > 2^{-1.48j}$ for every integer $j > 2$ with $j \notin \{3, 7, 8\}$.

Let $q \ge 10^5$. Lemma 11 gives $|\rho_i| \le 3/q$. Since $\sqrt{2} < 99/70$, we have $3\sqrt{2} < 297/70 < 4.3$, so $3\sqrt{2}/q < 4.3/q$. If $j \le 8$ and $|\rho_i| \le 3/q$, then $|x_i - 2^j\sqrt{2}| \le 3 \cdot 10^{-5} \cdot 2^8 \cdot 99/70 < 0.011$. Hence $|x_i^2 - 2^{2j+1}| < 0.011(2 \cdot 2^8 \cdot 99/70 + 0.011) < 8$, so $|x_i^2 - 2^{2j+1}| \le 7$. The element x_i is an odd integer at least 3. For any two distinct odd integers y and z at least 3, we have $|y^2 - z^2| \ge 16$. Thus, for this integer j, at most one odd integer $x \ge 3$ satisfies $|x^2 - 2^{2j+1}| \le 7$. Comparing 2^{2j+1} with those squares for $0 \le j \le 8$ leaves the pairs $(x_i, j) = (3, 0)$, $(3, 1)$, $(5, 2)$, $(11, 3)$ and $(181, 7)$. For 181 and $j = 7$, we have $2^{15} - 181^2 = 7$ and $2^{15/2} < 128 \cdot 99/70 < 181.03$, so

$$\left|181 \cdot 2^{-15/2} - 1\right| = \frac{7}{2^{15/2}(181 + 2^{15/2})} > \frac{7}{70000} = 10^{-4} > \frac{3}{q}.$$

For the other four pairs, $j \le 3$, $|x_i - 2^j\sqrt{2}| > 1/10$ and $2^j\sqrt{2} < 12$, so $|\rho_i| > 1/120 > 3/q$. Thus $j \ge 9$. Also $2^m = (q+\delta)/\sqrt{2} > 70q/99 > 2^{16}$, so $m \ge 17$ and $m+2 \ge 19$. Corollary 1.6 therefore applies to $(4q+1)/2^{m+2}$ and to $x_i/2^j$.

Since $2^m < (q+1)/\sqrt{2}$,

$$\left|\delta - \frac{1}{4}\right| > 2^m \cdot 2^{-1.48(m+2)} > 2^{-2.72}(q+1)^{-0.48}.$$

Also $\rho_i > -3/(q+1)$. Since $\sqrt{2} > 41/29$ and $q \ge 10^5$, we have $\sqrt{2}(1-3/(q+1)) > (41/29)(1-3/10^5)$. Here $41 \cdot 99997 > 141 \cdot 29 \cdot 1000$, so $(41/29)(1-3/10^5) > 1.41$, and therefore $2^j < x_i/1.41$. Combined with $\sqrt{2}\,|\rho_i| \le 3\sqrt{2}/q < 4.3/q$,

$$\frac{4.3}{q} > \Big(\frac{x_i}{1.41}\Big)^{-1.48},$$

so $x_i > 1.41\,(q/4.3)^{1/1.48}$. Lemma 11 gives $x_2 \ge q/3 - 1$. For $q > 93$ we have $q/3 - 1 > q/3.1$, so $1/x_2 < 3.1/q$. With $\sigma = 1/x_1 + 1/x_2 + 1/x_3$,

$$\frac{\sigma}{4} < \frac{4.3^{1/1.48}}{2 \cdot 1.41}\, q^{-1/1.48} + \frac{0.775}{q} < 0.951\, q^{-1/1.48} + \frac{0.775}{q},$$

where $4.3^{1/1.48}/(2 \cdot 1.41) < 0.951$.

Each $x_i \ge 3$, so $(q+\delta)^4 = (q+1)\prod_{i=1}^3 (q+1/x_i) \le (q+1)(q+1/3)^3$. The difference $(q+1/2)^4 - (q+1)(q+1/3)^3 = q^2/6 + 7q/54 + 11/432$ is positive, so $\delta < 1/2$. Write $\prod_{i=1}^3 (1+1/(qx_i)) = 1 + \sigma/q + \rho$, where ρ is the sum of the products of two or three factors $1/(qx_i)$. Then $0 < \rho \le 1/(3q^2) + 1/(27q^3)$, and Lemma 10 gives

$$\delta - \frac{1}{4} - \frac{\sigma}{4} = \frac{q\rho}{4}\Big(1 + \frac{1}{q}\Big) + \frac{\sigma}{4q} - \frac{3\delta^2}{2q} - \frac{\delta^3}{q^2} - \frac{\delta^4}{4q^3}.$$

Since $\sigma \le 1$ and $\delta < 1/2$, for $q \ge 10^5$ the absolute value of the right-hand side is at most

$$\frac{1+1/q}{12q} + \frac{1+1/q}{108q^2} + \frac{1}{4q} + \frac{3}{8q} + \frac{1}{8q^2} + \frac{1}{64q^3} < \frac{0.71}{q}.$$

Thus

$$\left|\delta - \frac{1}{4}\right| < 0.951\, q^{-1/1.48} + \frac{1.49}{q}.$$

At $q = 10^5$ we have $2^{-2.72}(q+1)^{-0.48} > 6.04 \times 10^{-4}$ and $0.951\, q^{-1/1.48} + 1.49/q < 4.13 \times 10^{-4}$. Dividing the two bounds by $(q+1)^{-0.48}$ leaves the constant $2^{-2.72}$ on the left. Each term on the right then has the form $c(q+1)^{0.48}q^{-a}$, with a positive real number c and a real number $a > 0.48$, so its logarithmic derivative $0.48/(q+1) - a/q$ is negative. The right side decreases for $q \ge 10^5$, and the inequality persists. Appendix E searches every $h \le 100{,}000$, and therefore every $q < 2^{25{,}000}$. □

Corollary 13. Let $\ell = 2^a$ with $a \ge 2$. Only finitely many odd q have a cycle of C of length ℓ with starting value 1 whose total number of halvings is $\equiv \ell/2 \pmod{\ell}$, and a bound is computable from Bauer and Bennett's Corollary 1.6 (2002). For $\ell = 4$ every such q is ruled out by Corollary 12.

Proof. Write $h = \ell m + \ell/2$. Every multiple of h is congruent to 0 or $\ell/2$ modulo ℓ, so for an integer t every $2^{t/\ell}$ that occurs is an integer power of 2 or equals $2^{1/2}$. Let s be an integer in $\{1, \ell\}$, let p be an integer, and let $j \geq 9$ be an integer. Let v be the integer in $\{0, a\}$ with $v = 0$ when $s = 1$ and $v = a$ when $s = \ell$. Since $\ell = 2^a$, we have $s \cdot 2^j = 2^{j+v}$. Then $j + v \geq 9$, so $j + v > 2$ and $j + v \notin \{3, 7, 8\}$. Bauer and Bennett's Corollary 1.6 (2002) therefore gives

$$\left|\sqrt{2} - \frac{p}{2^{j+v}}\right| > 2^{-1.48(j+v)} = 2^{-1.48v} \cdot 2^{-1.48j}.$$

Since $s \cdot 2^j = 2^{j+v}$ and $v \leq a$, we have $2^{-1.48v} \geq 2^{-1.48a}$, and

$$\left|\sqrt{2} - \frac{p}{s \cdot 2^j}\right| > 2^{-1.48a} \cdot 2^{-1.48j}.$$

Let r be the positive integer $\ell/2$. Then $2^{r/\ell} = 2^{1/2}$. The hypothesis of Theorem 2 therefore holds with $R = \{\ell/2\}$, the real number $\mu = 1.48 < \varphi$, the positive real number $c = 2^{-1.48a}$ and the positive integer $j_0 = 9$. The bound is then the Q in the proof of Theorem 2 for these values. The case $\ell = 4$ is Corollary 12. □

Corollary 14. Let $\ell = 3 \cdot 2^a$ with $a \geq 1$. Only finitely many odd q have a cycle of C of length ℓ with starting value 1 whose total number of halvings is $\equiv \ell/3$ or $2\ell/3 \pmod{\ell}$, and a bound is computable from Theorem (Bauer and Bennett (2018)). The length 3 is Theorem 1, which rules out every such q.

Proof. Write $h = \ell m + r_0$ with $r_0 \in \{\ell/3,\, 2\ell/3\}$. Every multiple of h is congruent to 0, $\ell/3$, or $2\ell/3$ modulo ℓ, so for an integer t the only irrational values of $2^{t/\ell}$ that occur are $2^{1/3}$ and $2^{2/3}$. In the proof of Theorem 2 the fraction is $(\ell q + 1)/(\ell \cdot 2^m)$. Its numerator is coprime to ℓ, and $\ell \cdot 2^m = 3 \cdot 2^{m+a}$, so this is a fraction of the shape in Theorem (Bauer and Bennett (2018)) with $s = 3$ and $j = m + a$. Each fraction $x_i/2^j$ has odd numerator x_i, so Theorem (Bauer and Bennett (2018)) applies with $s = 1$. Both give the exponent $\mu = 3\log_2(16166467234)/63 \approx 1.61487 < \varphi$ once j in Theorem (Bauer and Bennett (2018)) is at least 11000. The shift by a, and the constant factors in Theorem (Bauer and Bennett (2018)), contribute only a constant depending on ℓ. The fractions used in the proof of Theorem 2 therefore satisfy its hypothesis with $R = \{\ell/3,\, 2\ell/3\}$, this μ, a $c > 0$ read from Theorem (Bauer and Bennett (2018)), and $j_0 = 11000$. The bound is then the Q in the proof of Theorem 2 for these values. The length 3 is Theorem 1. □

7 Ineffective finiteness

Ridout's theorem (1958), S-unit form. Let α be an algebraic irrational, let S be a finite set of primes, and let $\varepsilon > 0$. Only finitely many reduced fractions p/Q with $Q > 0$ have every prime factor of Q in S and satisfy

$$\left|\alpha - \frac{p}{Q}\right| < Q^{-1-\varepsilon}.$$

This form gives no bound on the exceptional fractions. Since α is irrational, the finitely many fractions that meet it for a constant α lie at a positive distance from α, and these distances have a positive minimum. This is why Theorem 3, which uses this form in place of a restricted irrationality exponent, gives no bound on q.

Theorem 3 (finiteness for starting value 1). For each $\ell \geq 2$, only finitely many odd q have a cycle of C of length ℓ with starting value 1.

The argument below uses only the S-unit form of Ridout's theorem and the proof of Theorem 2. The constants supplied by Ridout's theorem are not explicit, so the resulting bound on q is not computed.

Proof. Let ε be a constant positive real number such that $1+\varepsilon < \varphi$. Let S be the finite set consisting of the prime 2 and the prime factors of ℓ. Let c be the constant positive real number $\ell^{-1-\varepsilon}$. For each integer r with $1 \leq r \leq \ell - 1$, the real number $2^{r/\ell}$ is an algebraic irrational. Ridout's theorem, applied to $2^{r/\ell}$ with this S and this ε, allows only finitely many reduced fractions p/Q, with $Q > 0$ and with every prime factor of Q in S, such that

$$\left|2^{r/\ell} - \frac{p}{Q}\right| < Q^{-1-\varepsilon}.$$

The same theorem with $\varepsilon/2$ in place of ε allows only finitely many reduced fractions with

$$\left|2^{r/\ell} - \frac{p}{Q}\right| < Q^{-1-\varepsilon/2}.$$

Every reduced fraction satisfies $\left|2^{r/\ell} - p/Q\right| > 0$. If $Q > 1$ and $\left|2^{r/\ell} - p/Q\right| \leq Q^{-1-\varepsilon}$, then $Q^{-1-\varepsilon} < Q^{-1-\varepsilon/2}$, so the fraction is one of those given by Ridout's theorem with $\varepsilon/2$. The reduced fractions with $Q = 1$ are integers p. Since $1 < 2^{r/\ell} < 2$, the inequality $\left|2^{r/\ell} - p\right| \leq 1$ holds for at most the integers $p = 1$ and $p = 2$. Thus only finitely many reduced fractions p_1/Q_1 satisfy $\left|2^{r/\ell} - p_1/Q_1\right| \leq Q_1^{-1-\varepsilon}$.

Let $p/(s \cdot 2^j)$ be a fraction with an integer p, with $s \in \{1, \ell\}$, and with an integer $j \geq 0$. Let p_1/Q_1 be that fraction written in lowest terms, so Q_1 is a positive integer, Q_1 divides $s \cdot 2^j$, and every prime factor of Q_1 lies in S. If $\left|2^{r/\ell} - p_1/Q_1\right| > Q_1^{-1-\varepsilon}$, then $Q_1 \leq s \cdot 2^j$ gives

$$\left|2^{r/\ell} - \frac{p}{s \cdot 2^j}\right| > Q_1^{-1-\varepsilon} \geq (s \cdot 2^j)^{-1-\varepsilon} = s^{-1-\varepsilon}\, 2^{-(1+\varepsilon)j} \geq c \cdot 2^{-(1+\varepsilon)j}.$$

The fractions p_1/Q_1 that remain, those with $\left|2^{r/\ell} - p_1/Q_1\right| \leq Q_1^{-1-\varepsilon}$, are finite in number for each r, and there are finitely many integers r with $1 \leq r \leq \ell - 1$. For each such fraction and each $s \in \{1, \ell\}$, the inequality $(s \cdot 2^j)^{-1-\varepsilon} < \left|2^{r/\ell} - p_1/Q_1\right|$ holds for every sufficiently large integer j. Hence there exists a constant integer $j_0 \geq 1$ such that, for every integer r with $1 \leq r \leq \ell - 1$, every integer p, every $s \in \{1, \ell\}$, and every integer $j \geq j_0$,

$$\left|2^{r/\ell} - \frac{p}{s \cdot 2^j}\right| > c \cdot 2^{-(1+\varepsilon)j}.$$

Ridout's theorem does not list the fractions p_1/Q_1, so this j_0 is not computed.

Let μ be the real number $1 + \varepsilon$. Then $\mu < \varphi$. The inequality displayed above is the inequality in the definition of a restricted irrationality exponent of $2^{r/\ell}$ for the set $\{1, \ell\}$, for every integer r with $1 \leq r \leq \ell - 1$. The constants c and j_0 are not explicit, so μ is not a restricted irrationality exponent in that definition. The proof of Theorem 2 uses only the inequality, with $R = \{1, \ldots, \ell - 1\}$ and with these values of μ, c, and j_0. That proof produces a real number Q. It does not compute Q. If $q > Q$, the same proof yields a contradiction. Hence $q \leq Q$. □

Theorem 3 fails for $\ell = 1$. A Mersenne number gives the cycle $1 \to 1$. Mathematically, for $\ell \geq 2$ an odd element other than 1 puts a factor strictly between q and $q+1$, so ℓ does not divide h and $2^{r/\ell}$ is irrational. Ridout's theorem then applies. For $\ell = 1$ that product has no factors and equals 1, so the same identity becomes $2^h = q + 1$, and the three conclusions of Lemma 10 fail. Here $2^{h/\ell} = 2^h$ is an integer, and Ridout's theorem does not apply.

Cycles with any starting value are treated in the companion paper [8]. The same finiteness, for a cycle with any starting value, is Theorem 2 of that paper.

8 Open problems

1. **Length 4 with h odd.** The δ window and the near misses for x_1 and x_3 involve $2^{1/4}$ and $2^{3/4}$. Write $h = 4m + r$ with an integer $m \geq 0$ and $r \in \{1, 3\}$. The denominator in the δ window is $4 \cdot 2^m = 2^{m+2}$. Theorem 2 would finish the case given a restricted irrationality exponent below φ of $2^{1/4}$ and of $2^{3/4}$, for the set $\{1, 4\}$. The construction of Bauer and Bennett (2002, 2018) needs a fourth power exceptionally close to a power of 2; among odd $a \leq 3 \times 10^6$ the best is $19^4 = 2^{17} - 751$, of quality $\log_2(2^{17})/\log_2 751 \approx 1.78$, against 4.4 for the cube case ($5^3 + 3 = 2^7$) and 5.3 for $\sqrt{2}$ ($181^2 + 7 = 2^{15}$). A different construction is likely needed.

2. **Effective results for larger ℓ.** For a general length ℓ, Theorem 2 needs μ to be a restricted irrationality exponent of $2^{r/\ell}$ for the set $\{1, \ell\}$, for every integer r with $1 \leq r \leq \ell - 1$. The case $S = \{1\}$ of that definition gives the denominators 2^j. Let $v \geq 0$ be an integer. When $\ell = 2^v$, we have $\ell \cdot 2^j = 2^{j+v}$, so that case also gives the denominators $\ell \cdot 2^j$, and $2^{-\mu v}$ is a constant real number. Santos's conjecture (2020) in full would need the hypothesis of Theorem 2 for every ℓ, or a bound uniform in ℓ.

A Cycles of length 1 and 2 with starting value 1

Proposition A.1 (length 1). The only cycle of length 1 has starting value 1 and $q = 2^h - 1$. In particular the cycle with starting value 1 is $1 \to 1$ precisely when q is a Mersenne number.

Proof. A cycle of length 1 is $C(x_0) = x_0$, so $x_0(2^h - q) = 1$. Both factors are positive integers, so $x_0 = 1$ and $2^h - q = 1$. Thus $q = 2^h - 1$. Conversely, if q is a Mersenne number, then $q = 2^e - 1$ for an integer $e \geq 2$, so $q \cdot 1 + 1 = 2^e$ and $C(1) = 1$. □

Proposition A.2 (length 2). The only odd q that has a cycle of length 2 with starting value 1 is $q = 5$.

Proof. The cycle gives $q(q+1) = 2^{k_1}(2^{k_2} - 1)$. Multiplying by 4 and adding 1,

$$(2q+1)^2 + (2^{k_1+2} - 1) = 2^{k_1+k_2+2}.$$

This is $X^2 + D = 2^n$ with $X = 2q + 1$ and $D = 2^{k_1+2} - 1$. By the theorem of Beukers (1980/81), stated as Theorem 1.8 of Bauer and Bennett (2002), for $D \neq 7$, 23 and D not of the form $2^k - 1$ there is at most one solution, while for $D = 2^k - 1$ with $k \geq 4$ there are exactly two, $X = 1$ and

$X = 2^{k-1} - 1$. These give $q = 0$ and $q = 2^{k_1} - 1$, the cycle $1 \to 1$ traversed twice, so $k_1 \geq 2$ gives nothing new. For $k_1 = 1$, $D = 7$ is the Ramanujan–Nagell equation with $X \in \{1, 3, 5, 11, 181\}$, and only $X = 11$ gives an odd q other than 1: $q = 5$, with $1 \to 3 \to 1$. □

As a check, the argument of Lemma B.2 below gives $2^{k_2} = qx + 1 < q^2 < 2^{2d}$ with d the order of 2 modulo q, so $k_2 = d$, and $q(q+1) = 2^{k_1}(2^d - 1)$; for $q = 5$, $d = 4$ and $5 \cdot 6 = 2 \cdot 15$.

B Earlier results on cycles of length 3 with starting value 1

These results were found before Theorem 1 and are not needed for it. Notation is as in Sections 3 to 5. Bauer and Bennett's headline inequality (2018), $|x^3 - 2^n| \geq x^{4/3}$ unless $x^3 = 2^n$ or $(x, n) = (5, 7)$, and their Corollary 4 (2018), $|x^3 - 27 \cdot 2^n| \geq 3^{5/3} x^{4/3}$ unless $x^3 = 27 \cdot 2^n$ or $x \in \{4, 5, 8, 15, 19, 38, 121\}$, are used throughout; both were also checked numerically for $n < 200$ and $n < 70$ respectively.

Factorization and the order of 2.

Lemma B.1 (factorization). With $q = 2^{k_1} x - 1$,

$$(2^{k_1} x - 1)(2^{k_1} x^2 - x + 1) = 2^{k_2}(2^{k_3} - 1).$$

Proof. The factors are q and $qx + 1$, and $q(qx + 1) = q \cdot 2^{k_2} y = 2^{k_2}(2^{k_3} - 1)$. □

Completing the square in $qx + 1 = 2^{k_2} y$ gives $(2q + 1)^2 + (2^{k_1+2} - 1) = 2^{k_1+k_2+2} \cdot y$, so a cycle of length 3 with starting value 1 gives the equation of Proposition A.2 with an odd cofactor y, tied to q by $qy + 1 = 2^{k_3}$.

Lemma B.2 (order of 2). $k_3 = d$ or $k_3 = 2d$, where d is the order of 2 modulo q. In particular $d < 3 \log_2 q$.

Proof. $2^{k_3} = qy + 1 \equiv 1 \pmod{q}$, so d divides k_3. Using $2^{k_1} \geq 2$ and $2^{k_2} \geq 2$, $y \leq (q^2 + q + 2)/4$, so $2^{k_3} \leq q(q^2 + q + 2)/4 + 1 < q^3$. Also $2^d \geq q + 1 > q$, so $q^3 < 2^{3d}$, and hence $k_3 < 3d$. □

Corollary B.3 (Wieferich primes). Let $F = (2^d - 1)/q$ and $k_3 = md$. Then $mF \equiv 2^{-k_2} \pmod{q}$, so $\gcd(F, q) = 1$. In particular no Wieferich prime q ($q^2 \mid 2^{q-1} - 1$) has a cycle of length 3 with starting value 1.

Proof. Since $2^d \equiv 1 \pmod{q}$, $y = (2^{md} - 1)/q = F(1 + 2^d + \cdots + 2^{(m-1)d}) \equiv mF \pmod{q}$. Reducing $2^{k_2} y = qx + 1 \bmod q$ gives $2^{k_2} \cdot mF \equiv 1$. For a Wieferich prime, the lifting-the-exponent lemma of Lucas (1878) shows that q^2 divides $2^d - 1$, since q^2 divides $2^{q-1} - 1$ and q does not divide $(q-1)/d$; so $q \mid F$. □

This is the mechanism Franco and Pomerance (1995) use for the question, posed by Crandall (1978), of when 1 is not a term of the orbit. Together with Lemma 3, Lemma B.2 determines

everything from q: $k_1 = v_2(q+1)$, h is the unique integer with $q^3 < 2^h < (q+1)^3$, $k_3 \in \{d, 2d\}$ and $k_2 = h - k_1 - k_3$.

Finiteness via Ridout's theorem.

Proposition B.4. Only finitely many odd q have a cycle of length 3 with starting value 1.

Proof. By Lemmas 8 and 5, $\delta - 1/3 < 2^{k_1+1}/(3q)$ with $2^{k_1} = (q+1)/x < (q+1)\sqrt{2/q}$, so $\delta - 1/3 < 0.96\, q^{-1/2}$ for $q \geq 71$. With $h = 3m + r$, $\beta = 2^{r/3}$, $P = 3q + 1$ and $Q = 3 \cdot 2^m$ this gives $0 < \beta - P/Q < 2.9\, q^{-1/2}/Q$, and since $Q < 2.39(q+1)$ we get $|\beta - P/Q| < 4.6\, Q^{-3/2} < Q^{-5/4}$ for $Q > 450$. The reduced fraction still has a denominator built from 2 and 3, so Ridout's theorem with $S = \{2, 3\}$ and $\varepsilon = 1/4$ allows only finitely many such fractions; infinitely many q would give infinitely many distinct ones, since the errors tend to 0 and β is irrational. □

This is the case $\ell = 3$ of Theorem 3, and Theorem 1 supersedes it. The power 3/2 in $|\beta - P/Q| < 4.6\, Q^{-3/2}$ comes from the lower bounds $x > \sqrt{q/2}$ and $y > x$; with only $x, y \geq 3$ the error is of order $1/Q$ and Ridout's theorem gives nothing. The result is ineffective, which is why Theorem 1 needs Theorem (Bauer and Bennett (2018)) instead.

Elementary special cases.

Proposition B.5 ($k_1 = k_2 = 1$). No cycle of length 3 with starting value 1 has $k_1 = k_2 = 1$. Hence $q \not\equiv 1 \pmod 8$.

Proof. Lemma 2 gives $2^h = q^3 + q^2 + 2q + 4 = (q+1)(q^2+2) + 2$. With $q + 1 = 2x$ this becomes

$$4x^3 - 4x^2 + 3x + 1 = 2^n, \qquad n = h - 1,$$

and $q^2 + 2 = 4x^2 - 4x + 3$ divides $2^n - 1$. Since q is odd, $q^2 + 2 \equiv 3 \pmod 8$. Every prime p dividing $q^2 + 2$ has -2 as a square mod p, so $p \equiv 1$ or $3 \pmod 8$, and at least one is $\equiv 3 \pmod 8$. For that p, 2 is not a square mod p, so the order of 2 modulo p is even and divides n. So n is even, and $(X, Y) = (4x,\ 4 \cdot 2^{n/2})$ is an integral point on

$$E\colon Y^2 = X^3 - 4X^2 + 12X + 16.$$

Using Sage (passagemath) with proofs enabled, E has rank 2 and trivial torsion, generated by $(0, 4)$ and $(1, 5)$ with saturation index 1, and its integral points are $X \in \{0, 1, 4, 9, 84\}$, computed by the elliptic-logarithm method. $X = 4$ gives $q = 1$; $X = 84$ gives $x = 21$ but $Y = 752 = 4 \cdot 188$, and 188 is not a power of 2; the others do not have the form $4x$ with x odd and at least 3. For the congruence: if $q \equiv 1 \pmod 8$, then $k_1 = 1$ and $x \equiv 1 \pmod 4$, so $t > k_1$ in Lemma 4, and its proof forces $k_2 = k_1 = 1$. □

For $q = 41$, $q^3 + q^2 + 2q + 4$ is twice a square, and $qy + 1$ is not a power of 2, since $41 \cdot 431 + 1 = 17{,}672 = 8 \cdot 47^2$.

Proposition B.6 ($k_1 = 1$, $k_2 = 2$). No cycle of length 3 with starting value 1 has $k_1 = 1$ and $k_2 = 2$. Hence $q \not\equiv 13 \pmod{16}$.

Proof. Suppose $k_1 = 1$ and $k_2 = 2$. Lemma 2 gives $2^h = q^3 + q^2 + 2q + 8$. Multiplying,

$$(q+2)(q^2 - q + 4) = q^3 - q^2 + 4q + 2q^2 - 2q + 8 = q^3 + q^2 + 2q + 8,$$

so $q + 2$ divides 2^h. Lemma 1 gives $q \geq 3 \cdot 2^{k_1} - 1$. With $k_1 = 1$ this is $q \geq 5$, and q is odd, so $q + 2$ is an odd integer at least 7. The only positive odd divisor of 2^h is 1.

For the congruence: if $q \equiv 13 \pmod{16}$, then $q + 1 \equiv 14 \pmod{16}$, so $v_2(q+1) = 1$. The first step is $q + 1 = 2^{k_1} x$ with x odd, so $k_1 = 1$ and $x = (q+1)/2 \equiv 7 \pmod 8$. The second step is $qx + 1 = 2^{k_2} y$. Substituting $q = 2x - 1$ gives $qx + 1 = 2x^2 - x + 1$. Since $x \equiv 7 \pmod 8$, we have $x \equiv -1 \pmod 8$, and

$$2x^2 - x + 1 \equiv 2(1) - (-1) + 1 = 4 \pmod 8.$$

Since $2x^2 - x + 1 \equiv 4 \pmod 8$, let m be the integer $(2x^2 - x + 1 - 4)/8$. Then $2x^2 - x + 1 = 4(2m + 1)$ with $2m + 1$ odd, so $v_2(qx + 1) = 2$. Since y is odd, $k_2 = 2$. Lemma 2 then gives $2^h = q^3 + q^2 + 2q + 8 = (q+2)(q^2 - q + 4)$, and Lemma 1 gives $q \geq 5$, so $q + 2$ is an odd integer at least 7 dividing 2^h. Hence $q \not\equiv 13 \pmod{16}$. □

Both are superseded by Corollary B.8, which rules out $k_1 = 1$ altogether.

Effective bounds on k_1.

Proposition B.7. In any cycle of length 3 with starting value 1, $2^{k_1} > q^{1/3}/2$.

Proof. Apply Bauer and Bennett's Corollary 4 (2018) with $x = 3q + 1$ and $n = h$. If $(3q+1)^3 = 27 \cdot 2^h$, then 3 divides $3q + 1$. Since $3q + 1 \equiv 1 \pmod 3$, we have $3 \nmid 3q + 1$. The only integers in $\{4, 5, 8, 15, 19, 38, 121\}$ congruent to 1 modulo 3 are 4, 19, and 121, and these give $q = 1$, $q = 6$, and $q = 40$. Here $q = 1$ fails $q \geq 3$, and $q = 6$ and $q = 40$ are even. For an odd integer $q \geq 3$, neither case occurs.

Lemma 8 gives $N = 27 \cdot 2^h - (3q+1)^3 > 0$, so

$$N = \left|(3q+1)^3 - 27 \cdot 2^h\right| \geq 3^{5/3}(3q+1)^{4/3}.$$

Since $3q + 1 > 3q$,

$$3^{5/3}(3q+1)^{4/3} > 3^{5/3}(3q)^{4/3} = 27\, q^{4/3}.$$

By Lemma 8, $N < 54 \cdot 2^{k_1} q$. Hence $54 \cdot 2^{k_1} q > 27\, q^{4/3}$, and therefore $2^{k_1} > q^{1/3}/2$. □

Corollary B.8. (a) $k_1 = 1$ is impossible: it would force $q < 64$, and the direct search excludes every $q < 2{,}000{,}000$; so every possible q is $\equiv 3 \pmod 4$. (b) Together with Lemma 5, $2^{2k_1 - 1} - 1 < q < 2^{3k_1 + 3}$, so each k_1 allows only an explicit finite range of q. (c) Since the digit scan (Appendix E) excludes $q < 2^{1{,}071{,}999{,}953}$, any remaining q would need $k_1 \geq 357{,}333{,}317$, so $q \equiv -1 \pmod{2^{357{,}333{,}317}}$.

Theorem 1 rules out these q altogether.

Proposition B.9 (upper bound on k_1). $2^{k_1} \le 3.2\,x^{2/3}$, and hence $2^{k_1} < 2.01(q+1)^{2/5}$.

Proof. By Lemma 9, $\left|x^3 - 2^f\right| \le 3.2\,x^2/2^{k_1}$ and $\left|x - 2^{f/3}\right| \le 1.6/2^{k_1} < 1$, so $x^3 \ne 2^f$. Bauer and Bennett's headline inequality (2018) gives $\left|x^3 - 2^f\right| \ge x^{4/3}$, and therefore $2^{3k_1}x^{4/3} \le 3.2 \cdot 2^{2k_1}x^2$. The exception $x = 5$ forces $q \le 39$ by Lemma 5. □

Together with Proposition B.7, $q^{1/3}/2 < 2^{k_1} < 2.01(q+1)^{2/5}$.

The three cases of Lemma 4.

Proposition B.10 ($t > k_1$ is impossible). No cycle of length 3 with starting value 1 has $t > k_1$.

Proof. Lemma 4 and its proof give $t = 2k_1$ and $k_2 = k_1$, so $x = 1+2^{2k_1}s$ with s odd. Proposition B.7 gives $q < 8 \cdot 2^{3k_1}$, so $x < 8 \cdot 2^{2k_1}$ and $s \in \{1, 3, 5, 7\}$. Then $q = s \cdot 2^{3k_1} + 2^{k_1} - 1$, and the cycle equation reads $2^h = q^3 + q^2 + 2^{k_1}q + 2^{2k_1}$. Dividing by 2^{9k_1},

$$s^3 < \frac{2^h}{2^{9k_1}} < \left(s + \frac{1}{2^{2k_1}}\right)^3 \left(1 + \frac{2}{q}\right),$$

and for $2^{k_1} \ge 8$ the right-hand side is below the next power of 2 after s^3 (2, 32, 128, 512), so $2^h/2^{9k_1}$ lies strictly between consecutive powers of 2. For $2^{k_1} \le 4$, $q \le 451$, which the direct search covers. □

(Also checked in exact arithmetic for $k_1 < 400$.)

Proposition B.11 ($t = k_1$ is impossible). No cycle of length 3 with starting value 1 has $t = k_1$.

Proof. Here $x = 1 + 2^{k_1}s$ with s odd, so $q = 2^{2k_1}s + 2^{k_1} - 1$. Let $g = h - 6k_1$. Expanding the cycle equation,

$$0 < 2^g - s^3 = \frac{R}{2^{6k_1}} \le \frac{6s^2}{2^{k_1}}, \qquad R = 3{\cdot}2^{4k_1}s^2(2^{k_1}{-}1) + 3{\cdot}2^{2k_1}s(2^{k_1}{-}1)^2 + (2^{k_1}{-}1)^3 + q^2 + 2^{k_1}q + 2^{k_1+k_2},$$

using $q < 2 \cdot 2^{2k_1}s$ and $2^{k_2} < q$ ($2^{k_1} \ge 4$). Bauer and Bennett's headline inequality (2018) gives $2^g - s^3 \ge s^{4/3}$, so $2^{k_1} \le 6\,s^{2/3}$, while Proposition B.7 gives $8 \cdot 2^{3k_1} > q > 2^{2k_1}s$, so $2^{k_1} > s/8$. Hence $s < 48^3$ and $2^{k_1} \le 13{,}824$, so $q < 2 \cdot 2^{2k_1}s < 5 \times 10^{13}$, which the computations cover. The exception $s = 5$ gives $q < 4 \times 10^6$ through Proposition B.9. □

Proposition B.12 (a constraint when $t < k_1$). Suppose $t < k_1$, so that $k_2 = t$, and write $x = 2^{k_1}u - (2^{k_2} - 1)$ with u odd. Then $2^{k_1}/2^{k_2} \le 3u^{2/3}$.

Proof. Put $\delta' = 2^{k_1+k_2} - 2^{k_1} + 1$, so $q = 2^{2k_1}u - \delta'$, and $g = h - 6k_1$. Expanding,

$$0 < u^3 - 2^g = \frac{3 \cdot 2^{4k_1}u^2\delta' - 3 \cdot 2^{2k_1}u\delta'^2 + \delta'^3 - q^2 - 2^{k_1}q - 2^{k_1+k_2}}{2^{6k_1}} \le \frac{3u^2 \cdot 2^{k_2}}{2^{k_1}},$$

since the terms after the first are negative in total and $\delta' < 2^{k_1+k_2}$. Bauer and Bennett's headline inequality (2018) gives $u^3 - 2^g \geq u^{4/3}$. The exceptions are $u = 1$, where $x \leq 2^{k_1}$ and Proposition B.9 forces $2^{k_1} \leq 32$, and $u = 5$, where it forces $q < 4 \times 10^6$. □

The positivity and upper bounds in Propositions B.11 and B.12 were also checked in exact arithmetic on hundreds of thousands of random parameter choices. Propositions B.10 and B.11 leave only the case $t < k_1$; Theorem 1 makes the case analysis unnecessary.

Small elements and an earlier bound on y.

Remark (small elements). The elements x and y are odd and different from 1, so $x, y \geq 3$, and Lemma 7 gives $y \geq 5$. If $x = 3$, Lemma 5 gives $2^{k_1} \leq 2(x-1) = 4$, so $q = 3 \cdot 2^{k_1} - 1 \in \{5, 11\}$, and neither has a cycle of length 3 with starting value 1. Hence $x \geq 5$ and $y \geq 7$. None of the proofs in the main text need more than $x \geq 3$ and $y > x$.

Before Lemma 6 was found, the case $y < x$ could not be excluded, and the following weaker bound was used instead. Lemma 7 shows that $y < x$ never happens, so the second alternative below is vacuous.

Proposition B.13. In a cycle of length 3 with starting value 1, either $y \geq x$ or

$$y > \left(\frac{3q^2}{q+1}\right)^{1/3} \approx (3q)^{1/3}.$$

Moreover, in the second case $v_2(y-1) = k_1$ exactly.

Proof. Multiplying $2^{k_2}y = qx + 1$ by $2^{k_1}y^2$ and using $qy = 2^{k_3} - 1$ and $q + 1 = 2^{k_1}x$ gives the exact identity

$$2^{2k_3} + (y-2)\,2^{k_3} + C = 2^{k_1+k_2}y^3, \qquad C = 2^{k_1}y^2 - y + 1.$$

If 2^{k_3} divides C, then $C \geq 2^{k_3} = qy + 1$, which simplifies to $2^{k_1}y \geq q + 1$, that is $y \geq x$. Otherwise the first two terms are divisible by a higher power of 2 than C, so $v_2(C) = k_1 + k_2$ and $2^{k_1+k_2} \leq C < 2^{k_1}y^2$. Since the left side exceeds $2^{2k_3} > q^2y^2$, this gives $q^2y^2 < 2^{k_1+k_2}y^3 < 2^{k_1}y^5$, so $q^2 < 2^{k_1}y^3 \leq (q+1)y^3/3$. For the last claim, the first step $q + 1 = 2^{k_1}x$ gives $q \equiv -1 \pmod{2^{k_1}}$. Lemma 1 gives $k_3 \geq k_1$, so $qy + 1 = 2^{k_3} \equiv 0 \pmod{2^{k_1}}$ and $y \equiv 1 \pmod{2^{k_1}}$. If $v_2(y-1) > k_1$, then $v_2(C) = k_1$, forcing $k_2 = 0$, which is impossible. □

For $q \geq 71$ the cube root is the smaller of the two bounds, so this gave $y > 1.436\,q^{1/3}$ unconditionally, and for $q >$ 2,000,000 it gave $y \geq 183$. The identity was also checked by computer algebra: substituting the cycle relations into both sides and expanding gives the same expression.

C Work on cycles of length 4 with starting value 1

The equations of the cycle. Write a cycle of length 4 with starting value 1 as $1 \to x \to y \to z \to 1$, with

$$q + 1 = 2^{k_1}x, \qquad qx + 1 = 2^{k_2}y, \qquad qy + 1 = 2^{k_3}z, \qquad qz + 1 = 2^{k_4}.$$

The next three lemmas are Lemma 1 and Lemma 5 for a cycle of length 4 with starting value 1, together with the gain from a larger x in $qz + 1 = 2^{k_4}$. They constrain k_4 and x, not y or z.

Lemma C.1 (lower bound for k_4). $k_4 \geq k_1 + 3$, and $z \equiv 1 \pmod{2^{k_1}}$.

Proof. The odd elements other than 1 are at least 3, so $x \geq 3$ and $z \geq 3$. Since $q + 1 = 2^{k_1}x$, we have $q + 1 \geq 3 \cdot 2^{k_1}$, that is $q \geq 3 \cdot 2^{k_1} - 1$. Since $2^{k_4} = qz + 1$,

$$2^{k_4} \geq (3 \cdot 2^{k_1} - 1) \cdot 3 + 1 = 9 \cdot 2^{k_1} - 2.$$

Since $2^{k_1} \geq 2$, we have $9 \cdot 2^{k_1} - 2 \geq 8 \cdot 2^{k_1} = 2^{k_1+3}$, and hence $k_4 \geq k_1 + 3$. From $q + 1 = 2^{k_1}x$ we also have $q \equiv -1 \pmod{2^{k_1}}$. Thus $qz + 1 \equiv 1 - z \pmod{2^{k_1}}$. But $qz + 1 = 2^{k_4}$ is divisible by 2^{k_1}, because $k_4 \geq k_1$, so $1 - z \equiv 0 \pmod{2^{k_1}}$, that is $z \equiv 1 \pmod{2^{k_1}}$. □

Lemma C.2. $2^{k_1} \leq 2(x - 1)$ and $x > \sqrt{q/2}$.

Proof. By Lemma C.1, $z \equiv 1 \pmod{2^{k_1}}$, so $qy + 1 = 2^{k_3}z$ gives $y \equiv 1 - 2^{k_3} \pmod{2^{k_1}}$. Let $t = v_2(x - 1)$ and $s = (x - 1)/2^t$. If $t \geq k_1$, then $2^{k_1} \leq x - 1$. Otherwise $k_2 = t$ and $y = 2^{k_1-t}x^2 - s \equiv 2^{k_1-t} - s \pmod{2^{k_1}}$, since $x^2 \equiv 1 \pmod{2^t}$. So $M = 2^{k_1-t} + 2^{k_3} - (s + 1)$ is a multiple of 2^{k_1}. If $M = 0$, then $2^{k_1-t} < s + 1$, so $2^{k_1} < 2^t(s + 1) \leq 2(x - 1)$. If $M \leq -2^{k_1}$, then $s + 1 > 2^{k_1}$, so $2^{k_1} \leq x$. If $M \geq 2^{k_1}$, then $2^{k_3} > 2^{k_1-1}$, so $k_3 \geq k_1$ and $2^{k_1-t} \equiv s + 1 \pmod{2^{k_1}}$, giving $2^{k_1-t} = s + 1$ or $s + 1 \geq 2^{k_1}$; either way $2^{k_1} \leq 2(x - 1)$. Since $q + 1 = 2^{k_1}x$,

$$q + 1 \leq 2x(x - 1) = 2x^2 - 2x.$$

Hence $q < 2x^2$, that is $x > \sqrt{q/2}$. □

Lemma C.3. $k_4 \geq 2k_1 + 1$ and $k_4 \geq 3k_1 - 1$.

Proof. By Lemma C.2, $x \geq 2^{k_1-1} + 1$. Since $z \geq 3$, we have $qz + 1 = 2^{k_4} \geq 3q + 1$. Since $q = 2^{k_1}x - 1$, we have $3q + 1 = 3 \cdot 2^{k_1}x - 2$, and therefore

$$2^{k_4} \geq 3 \cdot 2^{2k_1-1} + 3 \cdot 2^{k_1} - 2 > 2^{2k_1},$$

so $k_4 \geq 2k_1 + 1$.

By Lemma C.1, $z \equiv 1 \pmod{2^{k_1}}$ and $z \neq 1$, so $z \geq 1 + 2^{k_1}$. Then $2^{k_4} = qz + 1 \geq (2^{k_1}x - 1)(2^{k_1} + 1) + 1 = 2^{k_1}(x(2^{k_1} + 1) - 1)$. Substituting $x \geq 2^{k_1-1} + 1$ yields

$$x(2^{k_1} + 1) - 1 \geq (2^{k_1-1} + 1)(2^{k_1} + 1) - 1 = 2^{2k_1-1} + 2^{k_1-1} + 2^{k_1},$$

hence

$$2^{k_4} \geq 2^{3k_1-1} + 2^{2k_1-1} + 2^{2k_1} = 2^{3k_1-1} + 3 \cdot 2^{2k_1-1} > 2^{3k_1-1},$$

so $k_4 \geq 3k_1 - 1$. □

The middle elements.

Lemma C.4. If $k_2 \le k_3$, then either $y > q$, or $y > x$ and $y^2 > qx$. In particular $y \gg q^{3/4}$. If $k_3 \le k_4$, then either $z > q$, or $z > y$ and $z^2 > qy$.

Proof. Apply Lemma 6 with $i = 2$. Then $x_1 = x$, $x_2 = y$, and $x_3 = z$. If $k_2 \le k_3$ and $y < x$, then $y > q$; if $y > x$, then $y^2 > qx$. In the first case $y \gg q^{3/4}$. In the second, Lemma C.2 gives $x > \sqrt{q/2}$, so $y^2 > q^{3/2}/\sqrt{2}$ and again $y \gg q^{3/4}$. Apply Lemma 6 with $i = 3$. Then $x_2 = y$, $x_3 = z$, and $x_4 = x_0 = 1$. If $k_3 \le k_4$ and $z < y$, then $z > q$; if $z > y$, then $z^2 > qy$. □

Lemma C.4 bounds y only when $k_2 \le k_3$, and z only when $k_3 \le k_4$. It leaves open $k_3 < k_2$ with $y < x$, where y could be small, and $k_4 < k_3$, where z could be small. Those two lower bounds are the missing inputs for Proposition D.4. Theorem 2 of the companion paper [8] does not need them.

D Work on a cycle of any length with starting value 1

The structure of a cycle. The next three lemmas are Lemmas 1 to 3 for an arbitrary length.

Lemma D.1 (lower bound for k_ℓ). Let $\ell \ge 2$. On a cycle of length ℓ with starting value 1, $k_\ell \ge k_1 + 3$ and $x_{\ell-1} \equiv 1 \pmod{2^{k_1}}$.

Proof. Each of x_1 and $x_{\ell-1}$ is an odd element other than 1, so $x_1 \ge 3$ and $x_{\ell-1} \ge 3$. Since $q + 1 = 2^{k_1} x_1$, we have $q + 1 \ge 3 \cdot 2^{k_1}$ and $q \ge 3 \cdot 2^{k_1} - 1$. Also $2^{k_\ell} = q x_{\ell-1} + 1$. Since $q \ge 3 \cdot 2^{k_1} - 1$ and $x_{\ell-1} \ge 3$,

$$2^{k_\ell} = q x_{\ell-1} + 1 \ge (3 \cdot 2^{k_1} - 1) \cdot 3 + 1 = 9 \cdot 2^{k_1} - 2.$$

Since q is odd, $q + 1$ is even, so $k_1 \ge 1$ and $2^{k_1} \ge 2$. Then $9 \cdot 2^{k_1} - 2 - 8 \cdot 2^{k_1} = 2^{k_1} - 2 \ge 0$, and therefore $9 \cdot 2^{k_1} - 2 \ge 8 \cdot 2^{k_1} = 2^{k_1+3}$. Hence $2^{k_\ell} \ge 2^{k_1+3}$, so $k_\ell \ge k_1 + 3$.

From $q + 1 = 2^{k_1} x_1$, we have $q \equiv -1 \pmod{2^{k_1}}$. Since $k_\ell \ge k_1 + 3$, we have $k_\ell \ge k_1$, so $2^{k_\ell} \equiv 0 \pmod{2^{k_1}}$. Since $q x_{\ell-1} + 1 = 2^{k_\ell}$,

$$1 - x_{\ell-1} \equiv q x_{\ell-1} + 1 \equiv 2^{k_\ell} \equiv 0 \pmod{2^{k_1}}.$$

Hence $x_{\ell-1} \equiv 1 \pmod{2^{k_1}}$. □

The inequality $k_\ell \ge k_1 + 3$ is Lemma 1 when $\ell = 3$, and it is the first conclusion of Lemma C.1 when $\ell = 4$.

Lemma D.2 (cycle equation). For a cycle of length ℓ with starting value 1, set $S_0 = 0$. Then

$$2^h = q^\ell + \sum_{j=0}^{\ell-1} 2^{S_j} q^{\ell-1-j}.$$

Proof. For $1 \le m \le \ell$,

$$2^{S_m} x_m = q^m + \sum_{j=0}^{m-1} 2^{S_j} q^{m-1-j}.$$

The case $m = 1$ is $2^{k_1} x_1 = q + 1$. Suppose $m \geq 2$ and

$$2^{S_{m-1}} x_{m-1} = q^{m-1} + \sum_{j=0}^{m-2} 2^{S_j} q^{m-2-j}.$$

Since $2^{k_m} x_m = q x_{m-1} + 1$ and $S_m = S_{m-1} + k_m$,

$$2^{S_m} x_m = 2^{S_{m-1}} (q x_{m-1} + 1) = q \cdot 2^{S_{m-1}} x_{m-1} + 2^{S_{m-1}}.$$

Substituting the equation for $m - 1$ gives

$$2^{S_m} x_m = q \left(q^{m-1} + \sum_{j=0}^{m-2} 2^{S_j} q^{m-2-j} \right) + 2^{S_{m-1}} = q^m + \sum_{j=0}^{m-1} 2^{S_j} q^{m-1-j}.$$

At $m = \ell$, we have $x_\ell = 1$ and $S_\ell = h$, so the left side is 2^h and the right side is $q^\ell + \sum_{j=0}^{\ell-1} 2^{S_j} q^{\ell-1-j}$. □

Lemma 2 is the case $\ell = 3$.

Lemma D.3. For a cycle of length ℓ with starting value 1 and $\ell \geq 2$,

$$q^\ell < 2^h < (q+1)^\ell.$$

Hence ℓ does not divide h and $q = \lfloor 2^{h/\ell} \rfloor$.

Proof. Every term $2^{S_j} q^{\ell-1-j}$ in Lemma D.2 is positive, so $2^h > q^\ell$. Multiplying $2^{k_i} = (q x_{i-1} + 1)/x_i$ over $1 \leq i \leq \ell$ gives

$$2^h = \prod_{i=1}^{\ell} \frac{q x_{i-1} + 1}{x_i} = \frac{(q+1) \prod_{i=1}^{\ell-1} (q x_i + 1)}{(\prod_{i=1}^{\ell-1} x_i)\, x_\ell} = (q+1) \prod_{i=1}^{\ell-1} \left(q + \frac{1}{x_i} \right),$$

since $x_\ell = 1$. Each x_i with $1 \leq i \leq \ell - 1$ is an odd element other than 1, so $x_i \geq 3$ and $q + 1/x_i \leq q + 1/3 < q + 1$. Therefore $2^h < (q+1)^\ell$. A power of 2 strictly between consecutive ℓ-th powers is not an ℓ-th power, so $\ell \nmid h$. Thus $q < 2^{h/\ell} < q + 1$, and $q = \lfloor 2^{h/\ell} \rfloor$. □

Lemma 3 is the case $\ell = 3$. Lemma 10 includes these bounds.

Reduction to lower bounds. Before Theorem 3, finiteness for a cycle of length ℓ with starting value 1 was reduced to the hypothesis that every element other than 1 is at least $c \cdot q^\theta$ for constant real numbers $\theta > 0$ and $c > 0$.

Proposition D.4 (reduction to lower bounds). Let $\ell \geq 2$. Suppose there are constant real numbers $\theta > 0$ and $c > 0$ such that, on every cycle of length ℓ with starting value 1, every element other than 1 is at least $c \cdot q^\theta$. Then only finitely many q have a cycle of length ℓ with starting value 1.

Proof. On such a cycle, Lemma 10 gives $\ell \nmid h$. Write $h = \ell m + r$ with an integer $m \geq 0$ and an integer r satisfying $1 \leq r \leq \ell - 1$. Set $\beta = 2^{r/\ell}$, $P = \ell q + 1$ and $Q = \ell \cdot 2^m$. Then $2^m \beta = q + \delta$, so

$$\left|\beta - \frac{P}{Q}\right| = \frac{|\delta - 1/\ell|}{2^m}.$$

Lemma 10 gives $\delta - 1/\ell = \sigma/\ell + O(1/q)$. Thus there exist a constant real number $C > 0$ and a constant integer $q_0 \geq 3$ such that $|\delta - 1/\ell - \sigma/\ell| \leq C/q$ for every $q \geq q_0$. For every integer i with $1 \leq i \leq \ell - 1$, $x_i \geq c \cdot q^\theta$, so $\sigma \leq (\ell - 1)c^{-1}q^{-\theta}$. Let $\eta = \min(\theta, 1)$. Then η is a constant real number and $\eta > 0$. For $q \geq 1$, both $q^{-\theta} \leq q^{-\eta}$ and $q^{-1} \leq q^{-\eta}$, so

$$\left|\delta - \frac{1}{\ell}\right| \leq \left(\frac{\ell - 1}{\ell c} + C\right) q^{-\eta}$$

whenever $q \geq q_0$. Since $2^{1/\ell} \leq \beta < 2$ and $0 < \delta < 1$, we have $q/2 < 2^m < q + 1$. Hence $Q = \ell \cdot 2^m > \ell q/2$. For $q \geq 1$ we have $q + 1 \leq 2q$, so $Q < 2\ell q$ and $q^{-1-\eta} < (2\ell)^{1+\eta} Q^{-1-\eta}$. Let $K = 2((\ell - 1)/(\ell c) + C)(2\ell)^{1+\eta}$. Then K is a constant real number, and

$$\left|\beta - \frac{P}{Q}\right| < K\, Q^{-1-\eta}$$

for every $q \geq q_0$. Let p/Q_1 be the fraction P/Q written in lowest terms, with an integer $Q_1 > 0$. Every prime factor of Q_1 lies in the finite set S consisting of 2 and the prime factors of ℓ. Since $Q_1 \leq Q$ and $-1 - \eta < 0$,

$$\left|\beta - \frac{p}{Q_1}\right| < K\, Q_1^{-1-\eta}.$$

Let $\varepsilon = \eta/2$. Then ε is a constant real number and $\varepsilon > 0$. For every $Q_1 > K^{1/\varepsilon}$,

$$\left|\beta - \frac{p}{Q_1}\right| < Q_1^{-1-\varepsilon}.$$

Ridout's theorem, with this S and this ε, allows only finitely many fractions p/Q_1 satisfying $\left|\beta - p/Q_1\right| < Q_1^{-1-\varepsilon}$. The number β is irrational, so each of those fractions satisfies $\left|\beta - p/Q_1\right| > 0$. As $q \to \infty$, we have $|\beta - P/Q| < K\, Q^{-1-\eta} \to 0$, so each equals P/Q for only finitely many q. If instead $Q_1 \leq K^{1/\varepsilon}$, then Q_1 is a positive integer at most $K^{1/\varepsilon}$, so only finitely many such Q_1 occur. For each of them, only finitely many integers p satisfy $\left|\beta - p/Q_1\right| < 1$, and β irrational gives $\left|\beta - p/Q_1\right| > 0$. Let γ be the minimum of those finitely many positive real numbers $\left|\beta - p/Q_1\right|$. Then γ is a positive real number. For every q large enough that $K\, Q^{-1-\eta} < \gamma$, we have $\left|\beta - P/Q\right| < \gamma$, so $P/Q \neq p/Q_1$ for every such pair. Every odd integer $q < q_0$ is one of finitely many integers. Hence only finitely many q have a cycle of length ℓ with starting value 1. □

This is an earlier and weaker result than Theorem 3. It is conditional on the hypothesis that every element other than 1 is at least $c \cdot q^\theta$ for constant real numbers $\theta > 0$ and $c > 0$, whereas Theorem 3 is not.

E Computations

For a cycle of length ℓ with starting value 1 and $\ell \geq 2$, Lemma 10 gives $q^\ell < 2^h < (q + 1)^\ell$, so $\ell \nmid h$ and $q = \lfloor 2^{h/\ell} \rfloor$. Length 1 is Proposition A.1, and is not searched. A search of every $h \leq H$ with $\ell \nmid h$

therefore excludes every such cycle with $q < 2^{\lfloor H/\ell \rfloor}$: it would have $2^h < (q+1)^\ell \le 2^{\ell\lfloor H/\ell \rfloor} \le 2^H$, so $h \le H$. Searches for a cycle with any starting value, and searches at a constant q, are in the companion paper [8].

Cycles with starting value 1. For each ℓ and each $h \le H$ with $\ell \nmid h$, the candidate $q = \lfloor 2^{h/\ell} \rfloor$ is tested by applying C to 1, ℓ times. For $\ell = 3$, every $h \le 200{,}000$ was tested. Theorem 1 needs only $h < 66{,}003$: $q < 2^{22001}$ and Lemma 3 give $2^h < 2^{66{,}003}$. For each ℓ from 4 to 20, every $h \le 100{,}000$ was tested. No cycle with starting value 1 was found.

Cycles with starting value 1

length ℓ	h at most	no cycle with starting value 1 for q below
3	200,000	$2^{66{,}666}$
4	100,000	$2^{25{,}000}$
5	100,000	$2^{20{,}000}$
6	100,000	$2^{16{,}666}$
7	100,000	$2^{14{,}285}$
8	100,000	$2^{12{,}500}$
9	100,000	$2^{11{,}111}$
10	100,000	$2^{10{,}000}$
11	100,000	$2^{9{,}090}$
12	100,000	$2^{8{,}333}$
13	100,000	$2^{7{,}692}$
14	100,000	$2^{7{,}142}$
15	100,000	$2^{6{,}666}$
16	100,000	$2^{6{,}250}$
17	100,000	$2^{5{,}882}$
18	100,000	$2^{5{,}555}$
19	100,000	$2^{5{,}263}$
20	100,000	$2^{5{,}000}$

Checks of Lemmas 2, 4, 6, 8 and 9. The identities in these lemmas were checked by computer algebra, substituting the cycle relations and expanding both sides, or on random orbits. Lemma 4 was also checked numerically for every $k_1 \le 7$ and every odd $x < 20{,}000$. In the proof of Theorem 1, the two bounds on a were compared for every f from 33,000 to 400,000.

Digits of $2^{1/3}$ and $2^{2/3}$. A cycle of length 3 with starting value 1 forces places in the binary expansion of $2^{1/3}$ or of $2^{2/3}$ to alternate $0, 1, 0, 1, ...$, beginning with 0. Those places do not alternate in that way in the digits that were computed, except for $m < 100$, which the search over h already covers.

Proposition E.1 (digit scan). In a cycle of length 3 with starting value 1 and $q \ge 17$, write $h = 3m + r$ with $r \in \{1, 2\}$ and set $\beta = 2^{r/3}$. Then $q = \lfloor 2^m \beta \rfloor$. Places $m+1$ through $m + \lfloor m/2 \rfloor - 2$ in the binary expansion of the fractional part of β alternate $0, 1, 0, 1, ...$, beginning with 0. In particular, for $m \ge 100$, places $m + 1$ through $m + 48$ alternate in the same way.

Proof. Lemma 3 gives $q = \lfloor 2^{h/3} \rfloor$. Since $h = 3m + r$, we have $2^{h/3} = 2^m \beta$, so $q = \lfloor 2^m \beta \rfloor$. The fractional part of $2^m \beta$ is δ, and its binary digits are the digits of the fractional part of β from place

$m+1$ on. Lemma 8 gives $0 < \delta - 1/3 < 2^{k_1+1}/(3q)$; the upper bound uses $2^{k_1+k_2} \le 2^{k_1} q$ from Lemma 7. Since $q+1 = 2^{k_1}x$, and Lemma 5 gives $x > \sqrt{q/2}$, we have $2^{k_1} < (q+1)\sqrt{2/q}$ and

$$0 < \delta - \frac{1}{3} < \frac{2\sqrt{2}}{3}\left(1 + \frac{1}{q}\right) q^{-1/2}.$$

For $q \ge 17$ we have $1 + 1/q \le 18/17$, so $\frac{2\sqrt{2}}{3}(1+\frac{1}{q}) \le 12\sqrt{2}/17 < 1$, because $\sqrt{2} < 17/12$. Thus $\delta - 1/3 < q^{-1/2}$. Also $\beta > 1$, so $q = \lfloor 2^m \beta \rfloor \ge 2^m$ and $q^{-1/2} \le 2^{-m/2}$. Let $k = \lfloor m/2 \rfloor$. Then $m/2 \ge k$, so $2^{-m/2} \le 2^{-k}$, and

$$\frac{1}{3} < \delta < \frac{1}{3} + 2^{-k}.$$

In binary, $1/3 = 0.010101\ldots_2$: place $2j-1$ is 0 and place $2j$ is 1. Adding a positive amount less than 2^{-k} can change place k, and the carry stops at place $k-1$, because each 1 in this expansion is preceded by a 0. Places 1 through $k-2$ of δ therefore alternate $0, 1, 0, 1, \ldots$, beginning with 0. These are places $m+1$ through $m+k-2$ of the fractional part of β. For $m \ge 100$ we have $k-2 \ge 48$, so places $m+1$ through $m+48$ alternate in the same way. □

Let $N = 1{,}072{,}000{,}000$. The first N binary places of $\beta = 2^{1/3}$ and of $\beta = 2^{2/3}$ were computed and checked by $I^3 < 2^{3N+r} < (I+1)^3$, where $I = \lfloor 2^N \beta \rfloor$. A multiprecision value was kept only when its 192 guard digits were neither all 0 nor all 1 and its leading 61 digits matched an integer cube root. Above about 2^{30} digits that value disagreed with the integer test, so N was kept below 2^{30} and the integer test is the one used here. For $m \ge 100$, Proposition E.1 requires places $m+1$ through $m+48$ to alternate. Places $m+1$ through $m+48$ do not alternate in that way for any m with $m+48 \le N$, that is $m \le N-48$. Such an m would give $q = \lfloor 2^m \beta \rfloor < 2^{m+1} \le 2^{N-47} = 2^{1{,}071{,}999{,}953}$, since $\beta < 2$. Every $m < 100$ has $h = 3m + r < 300$, which the search $h \le 200{,}000$ includes. Together these rule out every cycle of length 3 with starting value 1 and $q < 2^{1{,}071{,}999{,}953}$.

Checking Bauer and Bennett (2018). The results of this paper rely heavily on Bauer and Bennett (2018), so the argument that produces their inequality was checked. The labels below are those of the published paper. In the preprint of Bauer and Bennett (2018) the same results are Theorem 3, inequality (21), Lemma 7, Proposition 8 and Lemma 9.

Theorem (Bauer and Bennett (2018)) is the inequality in the proof of their Theorem 2.1, before the exponent is rounded. The constant 16166467234 in that inequality is the integer just above $9 \cdot 2^{28} \cdot F(\tau)/1.79$, where F is their auxiliary function: $F(\tau) = 11.97804\ldots$ at $\tau = 0.20305\ldots$, and the product equals $16166467233.3\ldots$. The denominator of their inequality (5.3) contributes $1/(1.26 \cdot 2^{14/3}) = 2^{-5.00009\ldots}$, and the formula they display omits this factor. The factor 2^{-5} in Theorem (Bauer and Bennett (2018)) is a weakening of the contour bound for $|P_{n_1,\delta}|$, which has constant $1.2528\ldots$ in place of 1.26: then $1.2528\ldots \cdot 2^{14/3}$ has base-2 logarithm $4.9918 < 5$. The sentence before the proof of Theorem 1 records the same comparison, and the factor is absorbed in their final bound $2^{-1.62k}$.

Their Lemma 4.1 was checked at 3000-digit precision for every parameter n_1 from 2 to 60 and both values of each of δ and r. The inequality uses $n_1 \ge 497$, and Lemma 4.1 is uniform in n_1. With

$$f_{\delta,r}(v) = \frac{v^{4+r/3-\delta}(1-v)^{1-r/3}}{\left(1 - 3(1-v)/128\right)^{5-\delta}},$$

we have $\max f_{\delta,r} = 0.31999\ldots < 1/3$, and $F(\tau) < 12$.

Their Proposition 5.1 states that a product Π, built from their Padé approximants, satisfies $\Pi > 1.8^{n_1}$ for every integer $n_1 \geq 497$. It was recomputed as follows.

- Their Lemma 6.1 lists the primes that divide every Padé coefficient. For every $n_1 \leq 160$ and both values of each of δ and r, the 5,978 primes that meet its hypotheses were checked, with no failure.

- A lower bound $L(r, n_1)$ for $\log \Pi$ was computed exactly for every $497 \leq n_1 < 10^6$ and both values of r. It lower-bounds $\log \Pi$ through Lemma 6.1. The cutoff $\lfloor \sqrt{n_1}/12 \rfloor - 2$ in that application is negative for $n_1 < 576$, so on $n_1 \leq 830$ the sum includes primes beyond the cutoff, each still required to satisfy $p > \max\{\sqrt{3n_2 + 2},\, 5\}$. It reproduces their values $\exp(L(2, 3319)/3319) = 1.897735\ldots$ and $\exp(L(1, 10000)/10000) = 2.006523\ldots$. The bound exceeds $\log 1.8$ except for 43 pairs (n_1, r), all with $n_1 \leq 830$. For those pairs Π was computed exactly for both values of δ, and the minimum of Π^{1/n_1} is 1.80767, at $n_1 = 497$. For $1000 < n_1 < 10^6$ the minimum of $\exp(L(r, n_1)/n_1)$ is 1.8422.

- For $n_1 \geq 10^6$ the analytic lower estimates of $\log \Pi$ in the preprint of Bauer and Bennett (2018) were recomputed: $0.59197\, n_1$ when $10^6 \leq n_1 \leq 2 \times 10^9$, and $0.59035\, n_1$ when $n_1 > 2 \times 10^9$. Both exceed $\log 1.8 \cdot n_1 = 0.58779\, n_1$. Those estimates use the preprint's prime-error bound, in which $1.798158\sqrt{x}$ and $0.00144\, x/\log x$ meet at $x = 641{,}239{,}200.2$. The journal states the coefficient 0.001 and splits this argument at 3×10^9. The constant 1.798158, from Bennett, Martin, O'Bryant and Rechnitzer (2018), was spot-checked for $x \leq 2 \times 10^8$, and the largest ratio found is 1.79628.

What was not recomputed is that prime bound for $x > 2 \times 10^8$, Lemma 4 of Beukers (1980/81), where one choice of δ gives $\Lambda \neq 0$, and Lemma 4.5 of Chudnovsky (1983), which Lemma 6.1 uses.

F Notes on the literature and history

Literature.

- Santos (2020) studies cycles of C with starting value 1, the same question as this paper. Propositions 15 and 16 there are the cases of length 1 and length 2, and Conjecture 3 is that a cycle of C has starting value 1 only when $q = 5$ or when q is a Mersenne number. His Remark 5 describes a cycle of length 3 or more as "… much more difficult to analyze and seems intractable"; Theorem 1 is the case of length 3. The same preprint claims a proof of the Collatz conjecture, which is not accepted.

- Crandall (1978) and Franco and Pomerance (1995) study when 1 is not a term of an orbit of a $qx + 1$ map; Corollary B.3 matches their mechanism.

- Steiner (1977, 1981) rules out one-block "circuits" for $q = 3$, 5, 7 with Baker's method, a related use of Diophantine approximation for cycles.

- The deep inputs are Bauer and Bennett (2002, 2018) on restricted irrationality exponents,

Beukers (1980/81) on $X^2 + D = 2^n$, and Ridout (1958) on approximation with S-unit denominators.

History of the results. The work began with the δ window (Lemma 8), which pins down q from h and makes a search over h far faster than a search over q. Lower bounds on x and y (Lemmas 4 to 7, preceded by the weaker Proposition B.13, which allowed $y < x$ with $y > (3q)^{1/3}$) then gave finiteness through Ridout's theorem (Proposition B.4), but no bound. Residue-class arguments settled special cases (Propositions B.5 and B.6), and the effective inequality of Bauer and Bennett (2018) gave the lower bound on k_1 (Proposition B.7) and most of the case analysis of Lemma 4 (Appendix B). Playing the δ window against the near miss for x then gave Theorem 1, once it was noticed that the unrounded exponent of Bauer and Bennett (2018) lies below the golden ratio. Lemma 11 extended this to Theorem 2, Corollaries 12 to 14, and Theorem 3. The digit scan (Appendix E) was computational evidence found along the way. Cycles with any starting value are treated in the companion paper [8]. The final verification found and corrected a dropped constant in a displayed formula of Bauer and Bennett (2018) and confirmed their Proposition 5.1 for every $n_1 \geq 497$.

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